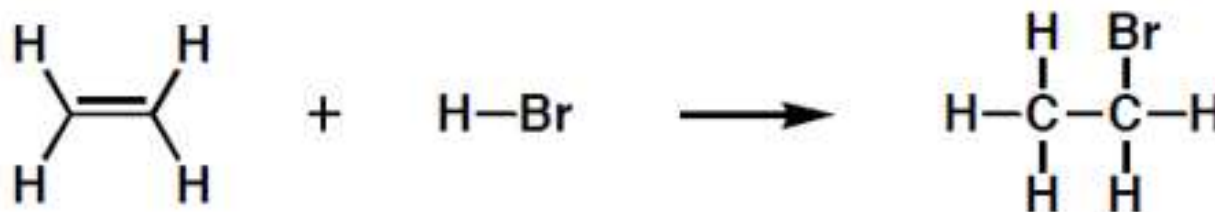


UNIT 4

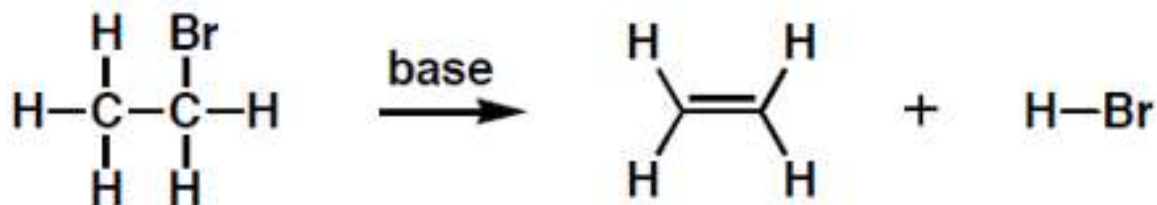
MAJOR ORGANIC REACTIONS

Kind of organic reaction

a. Addition



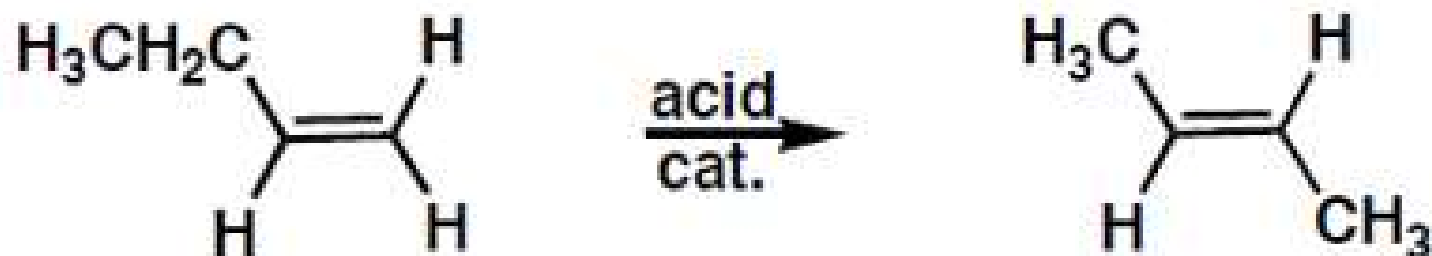
b. Elimination



c. Substitution



d. Rearrangement



Curved arrows indicate breaking and forming of bonds

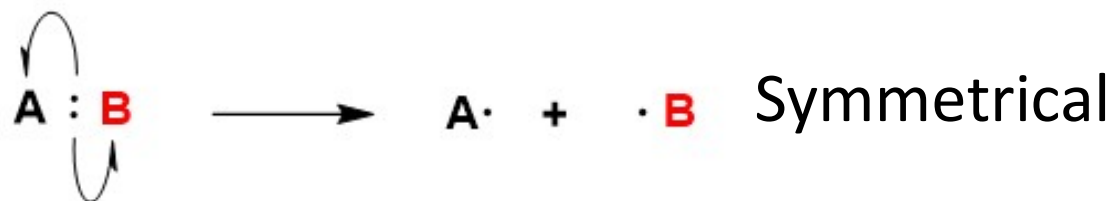


Arrowheads with a “half” head (“fish-hook”) indicate homolytic and homogenic steps



Arrowheads with a complete head indicate heterolytic and heterogenic steps

Bond breaking



Bond making



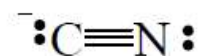
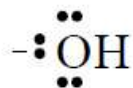
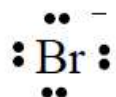
Nucleophile (Nu): - is a substance that is nucleus-loving.

has a negatively polarized, electron rich atom and can form a bond by donating electrons.

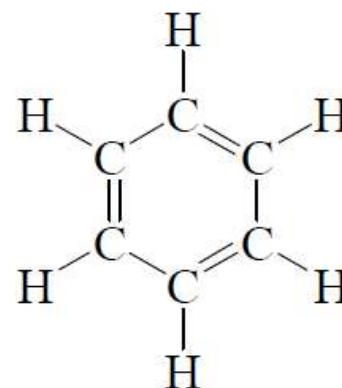
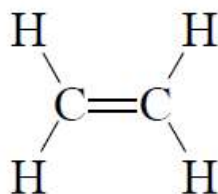
Characteristics: Nucleophilic atoms have either lone pairs or pi bonds that can be used to form new bonds to electrophiles

Nucleophile Examples

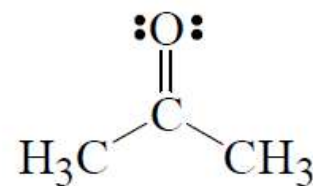
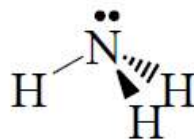
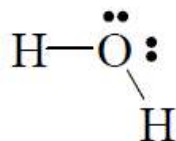
Anions



Pi bonds



Atoms with lone pairs



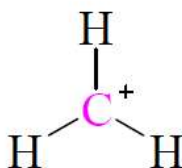
Electrophile (E): - is a substance that is electron-loving.

has a positively charge, electron poor atom and
can form a bond by accepting a pair of electrons
from nucleophiles.

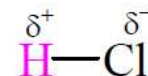
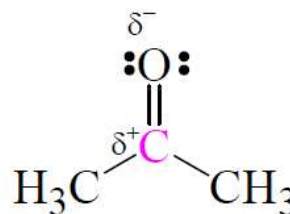
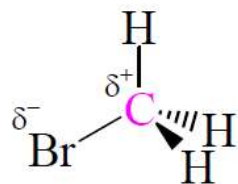
Characteristics: Electrophilic atoms have Positive charge, a partial positive charge, or be very polarizable

Electrophile Examples

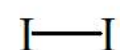
- Charged:



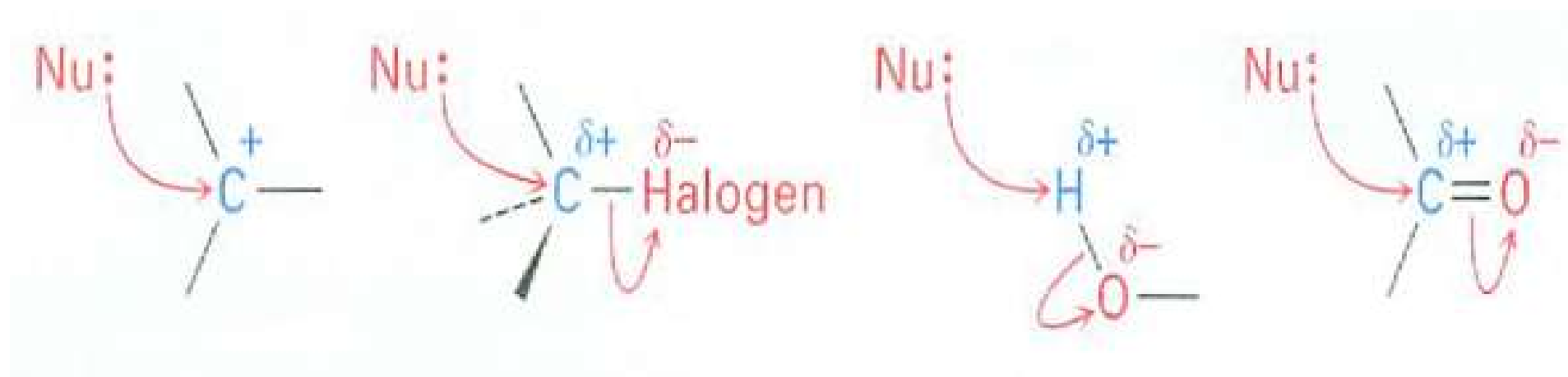
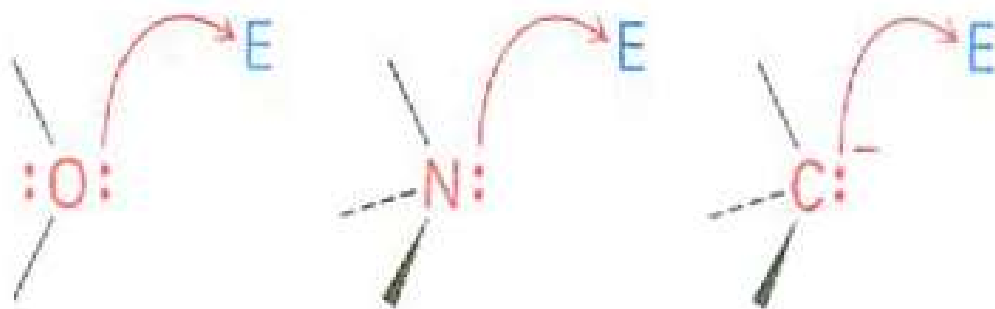
- Polar:

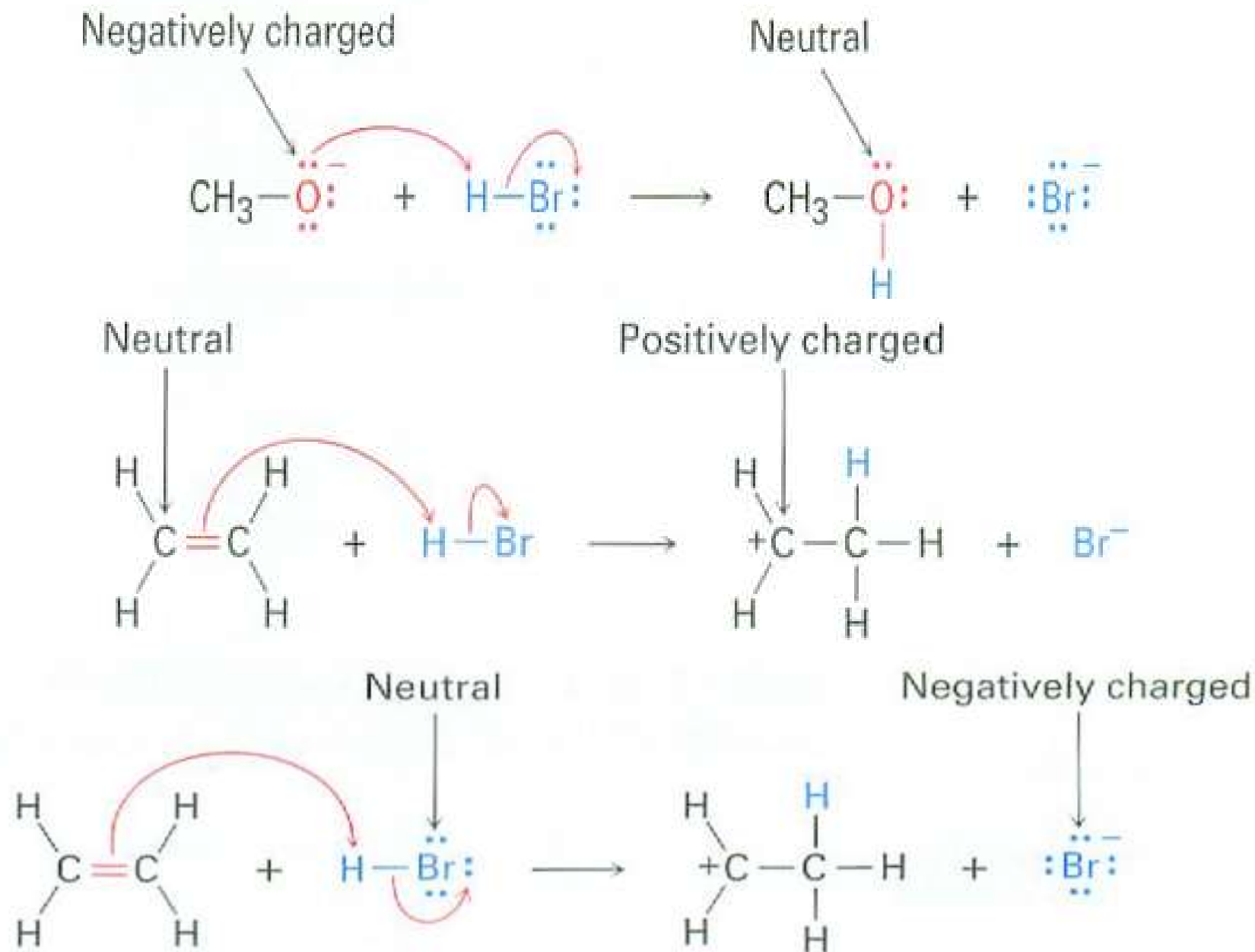


- Polarizable:



Electrons move from a nucleophilic source (Nu) to an electrophilic sink (E)

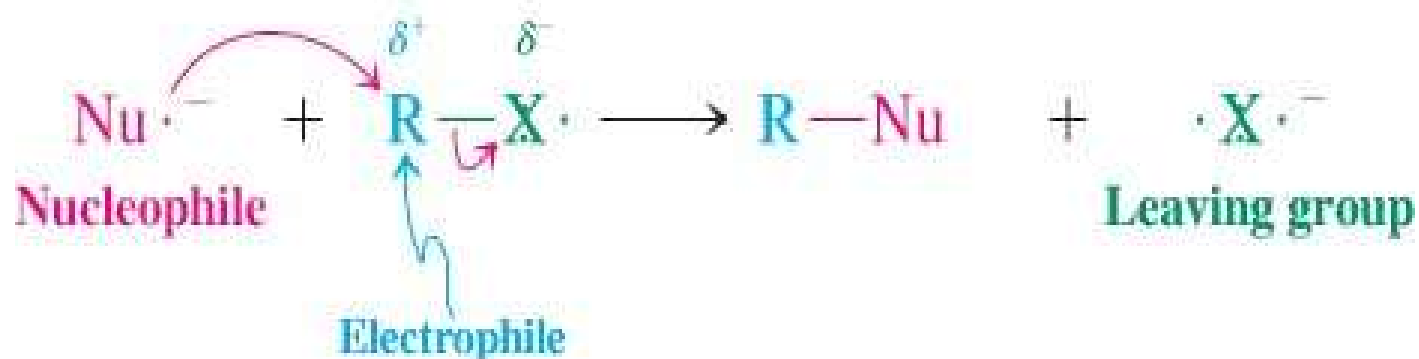




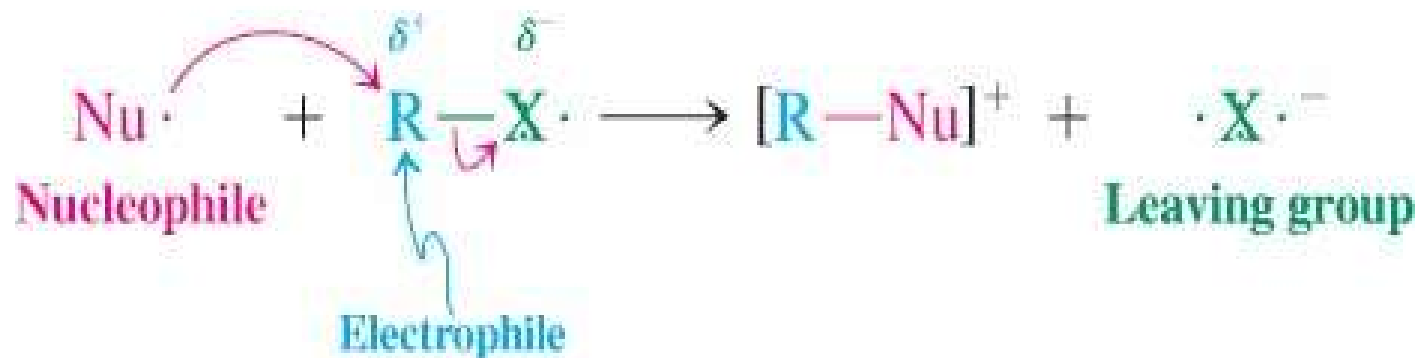
The octet rule must be followed during electron movement.

4.1 Substitution reactions

Nucleophilic Substitutions



Nucleophiles attack electrophilic centers



There are two mechanisms of substitution: S_N1 & S_N2

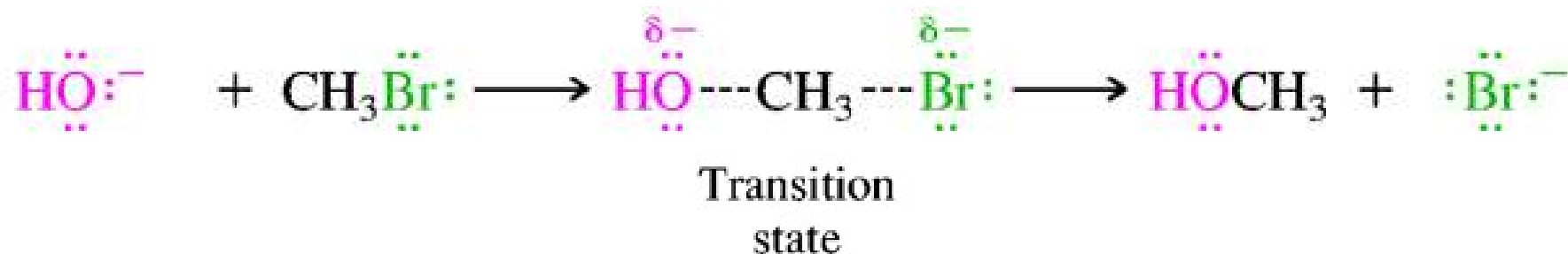
S_N2 Mechanism

- substitution **nucleophilic bimolecular**

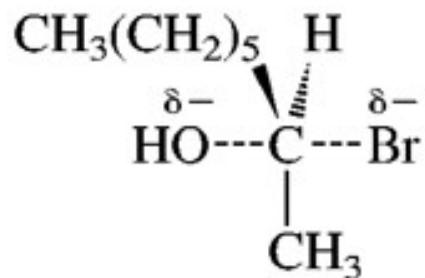
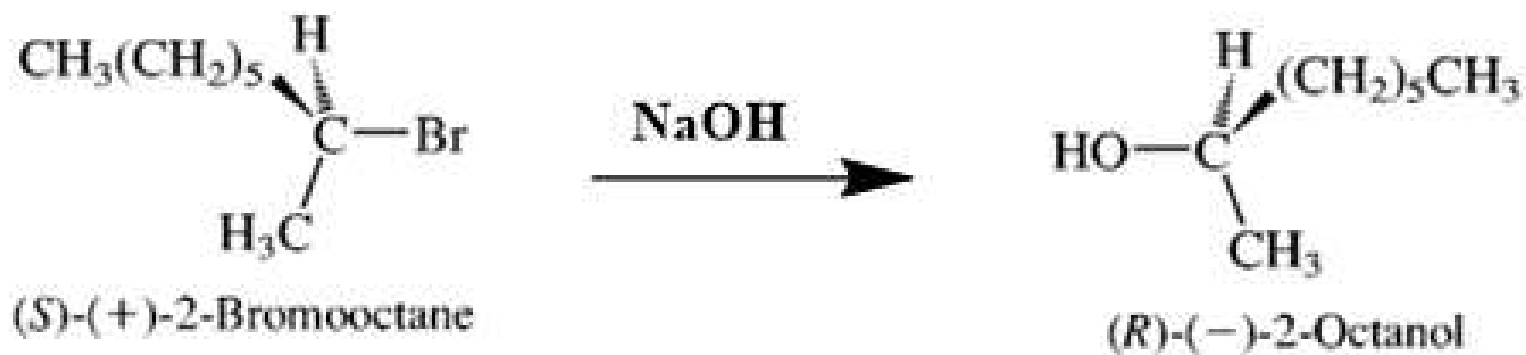
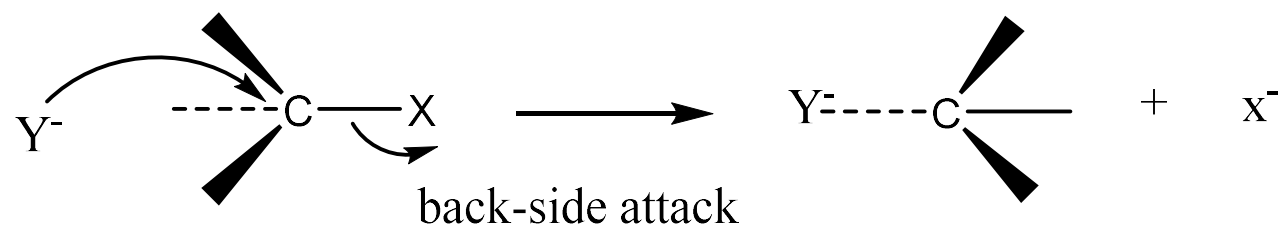


$$\text{Rate} = k[\text{CH}_3\text{Br}][\text{HO}^-] \quad \text{second order overall}$$

- It is a single-step process (concerted reaction) in which both the alkyl halide and the nucleophile are involved at the transition state.



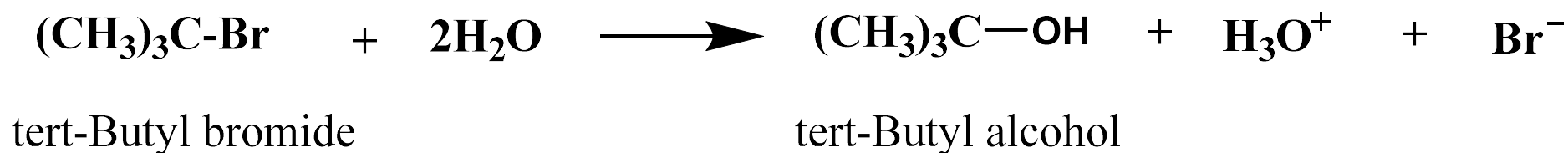
- S_N2 occurs with inversion of configuration with back side attack.



transition state

S_N1 Mechanism

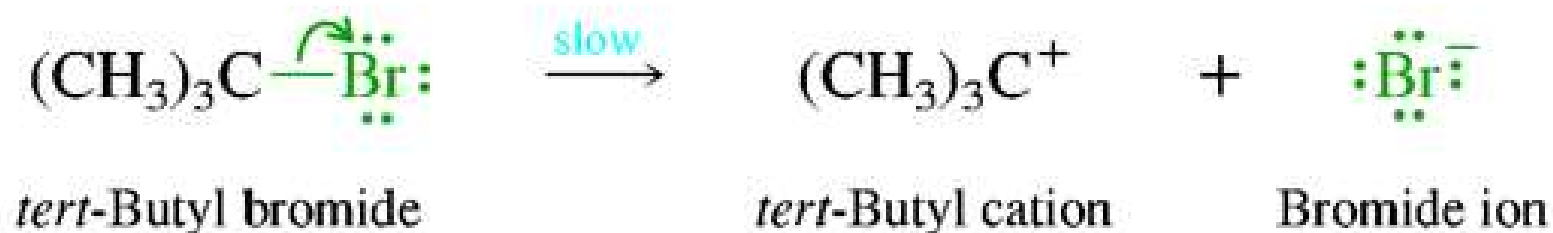
is Substitution nucleophilic unimolecular



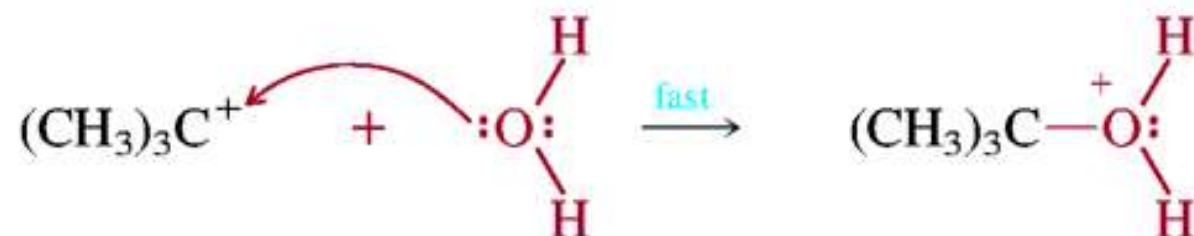
$$\text{Rate} = k[(\text{CH}_3)_3\text{CBr}] \quad \text{First order overall}$$

The rate depends on only the alkyl halide

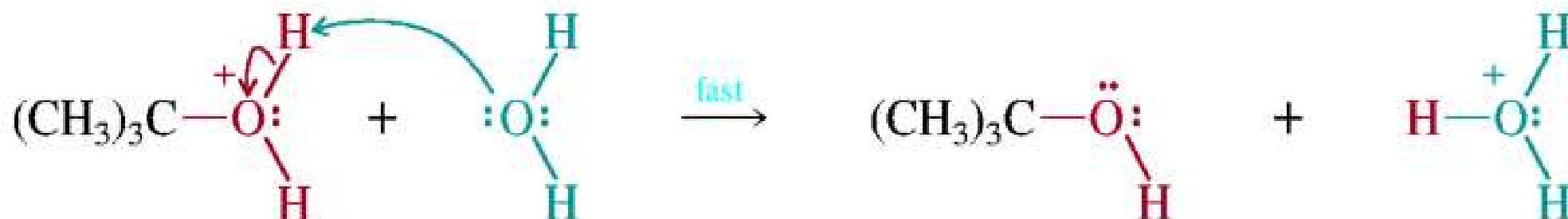
Step 1: The alkyl halide dissociates to a carbocation and a halide ion.



Step 2: The carbocation formed in step 1 reacts rapidly with a water molecule.



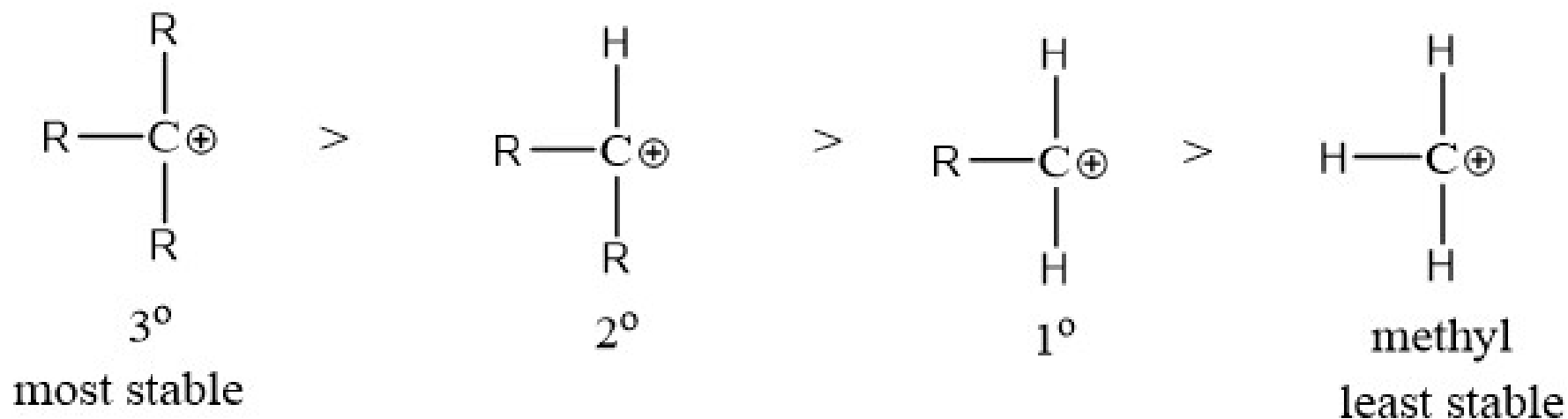
Step 3: This step is a fast acid-base reaction that follows the nucleophilic substitution. Water acts as a base to remove a proton from the alkyloxonium ion to give the observed product of the reaction, *tert*-butyl alcohol.



Carbocations are formed in SN1 reaction. Thus, the rate depends on the stability of carbocations.

Carbocation is a carbon cation $\begin{array}{c} + \\ \text{R}-\text{C}-\text{R} \\ | \\ \text{R} \end{array}$

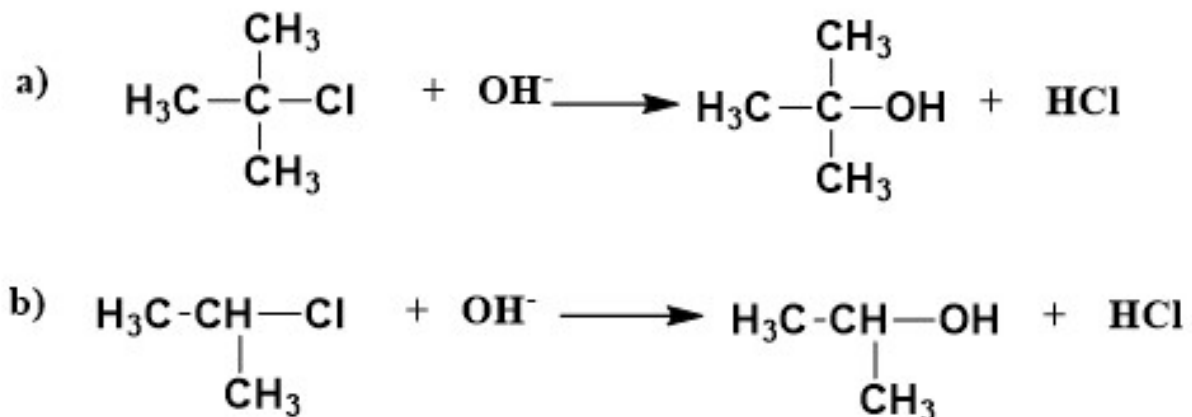
- The order of carbocation stability is: $3^\circ > 2^\circ > 1^\circ > \text{methyl}$



- Alkyl group stabilize carbocations by inductive effect.

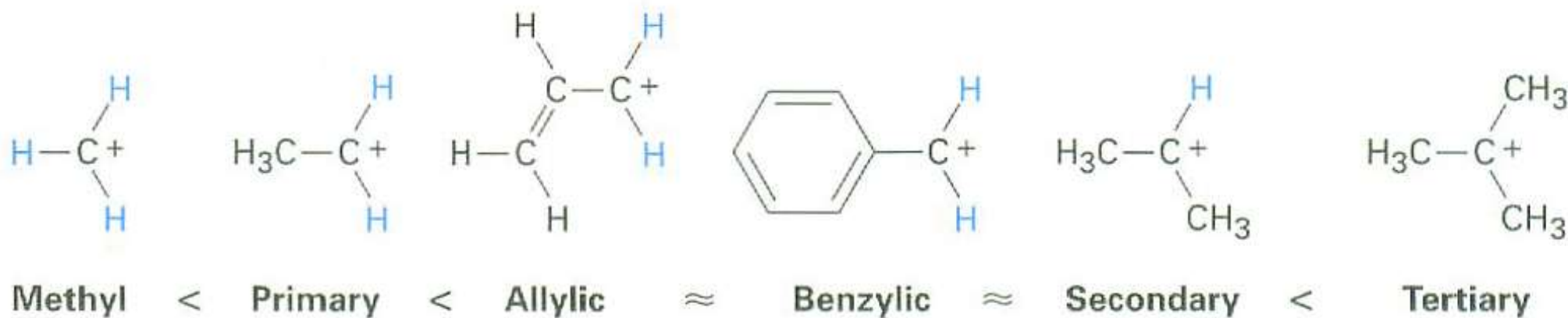


The more stable the carbocation, the rapid the reaction by SN1



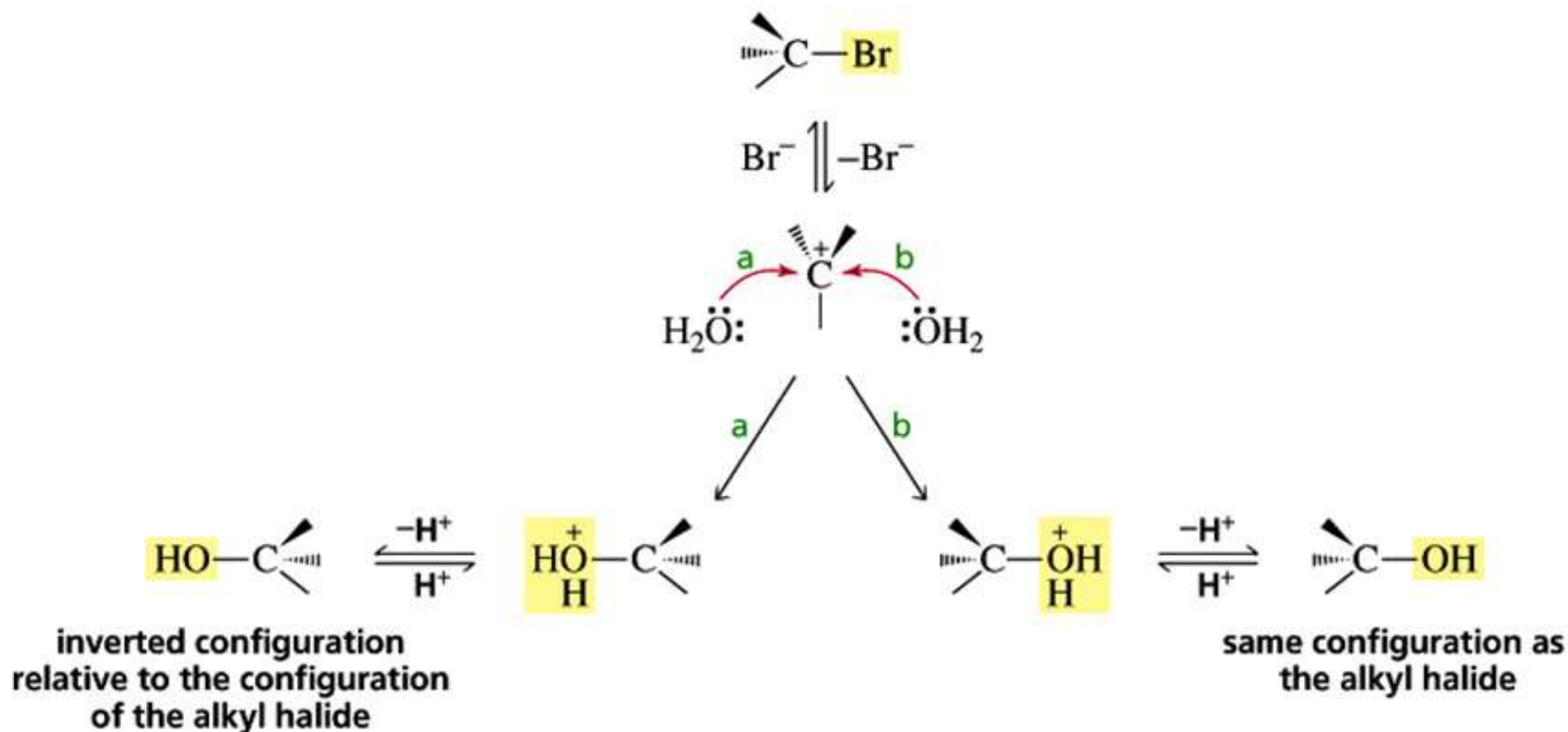
Reaction (a) takes place more rapidly than (b) since passes through more stable 3° carbocation.

Carbocation stability

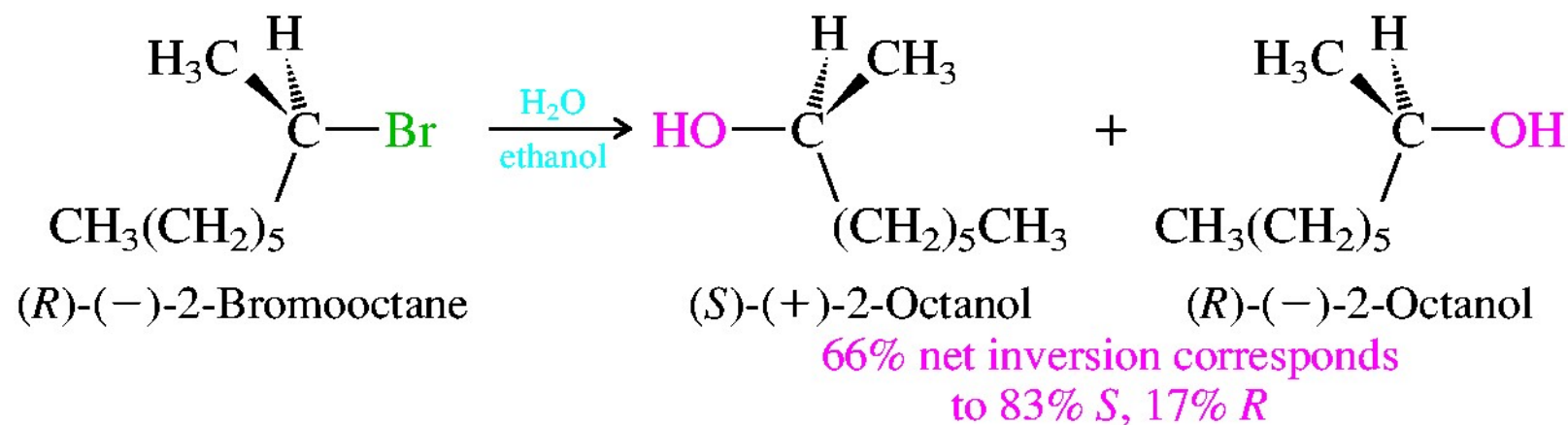


STEREOCHEMISTRY OF S_N1 REACTIONS

- S_N1 is not stereospecific
- Since carbocation has a trigonal planar structure, the Nucleophile can attack from front side or back side



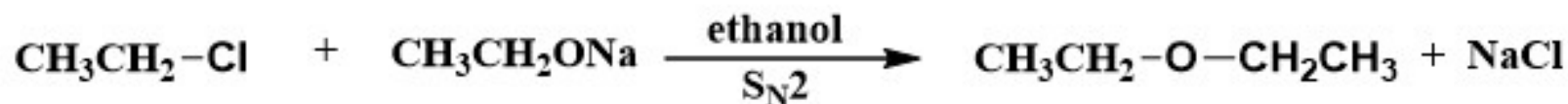
A mixture of (R) and (S) products are possible= Racemization



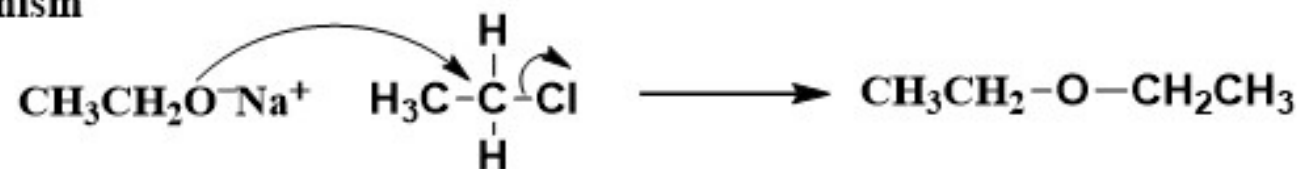
Applications of Substitution Reactions

1. Alkyl halides

- Williamson ether synthesis



mechanism



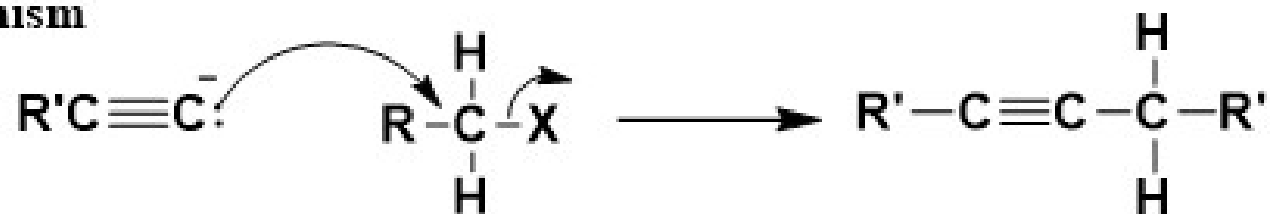
- preparation of primary amine



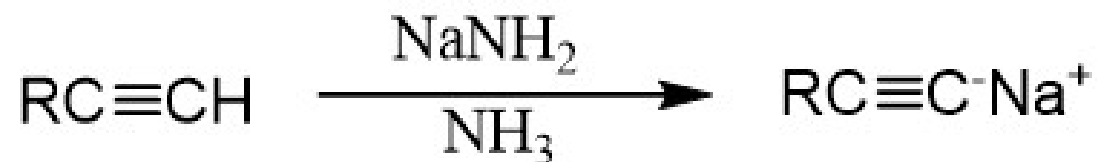
- Preparation of alkynes

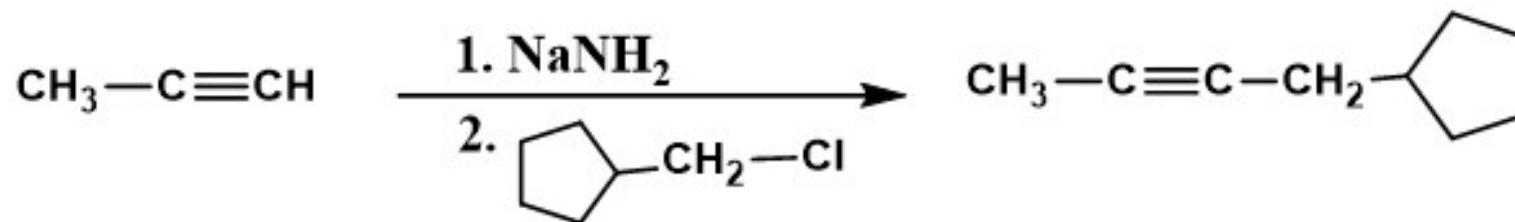


mechanism



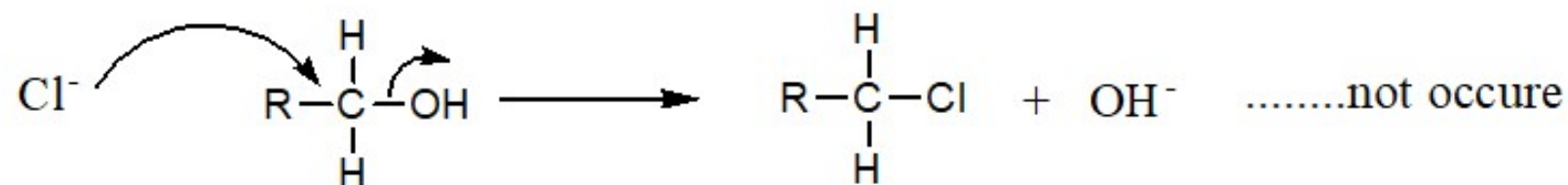
$\text{RC}\equiv\text{CNa}$ can be prepared from a terminal alkyne using strong base like NaNH_2



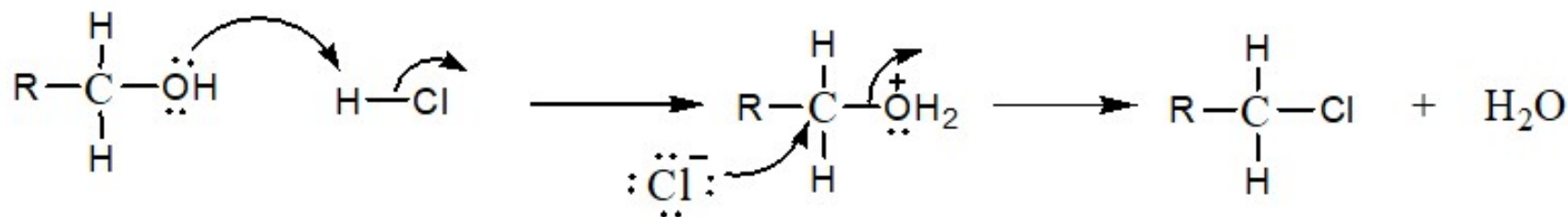


2. Alcohols

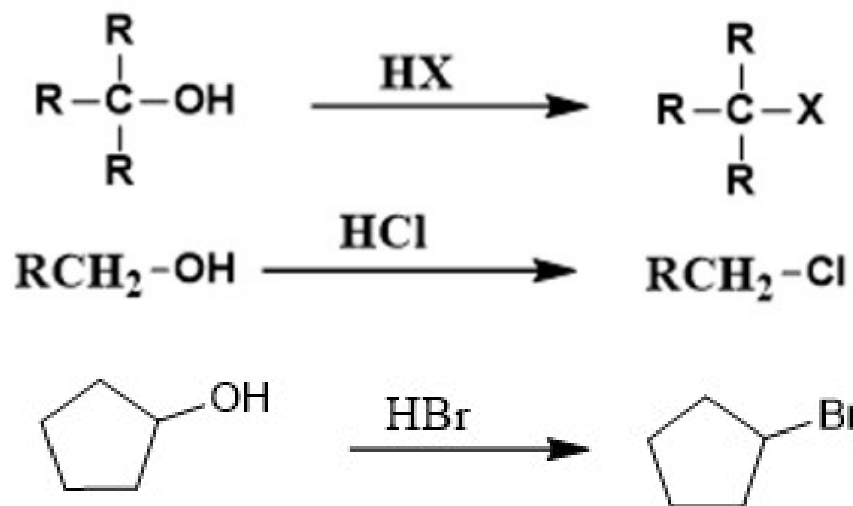
-OH is too basic to be displaced by a nucleophile



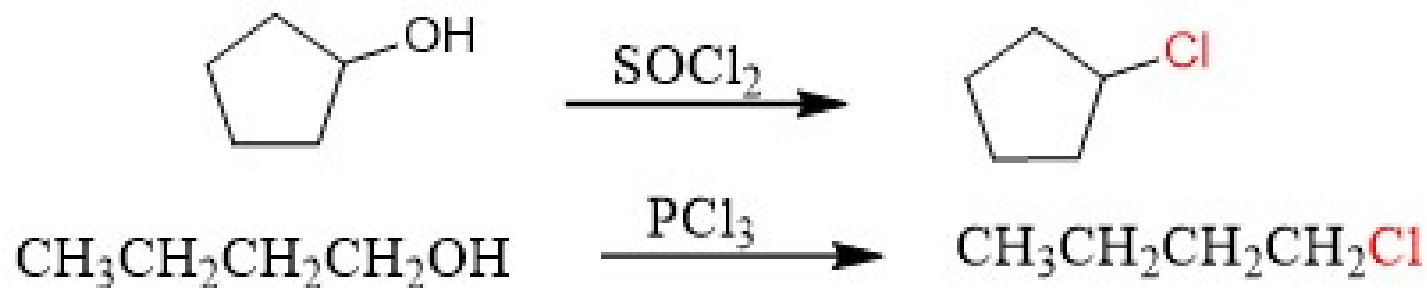
- Acid protonation change -OH to -OH₂⁺ (good leaving group)



Preparation of alkyl halide from alcohol



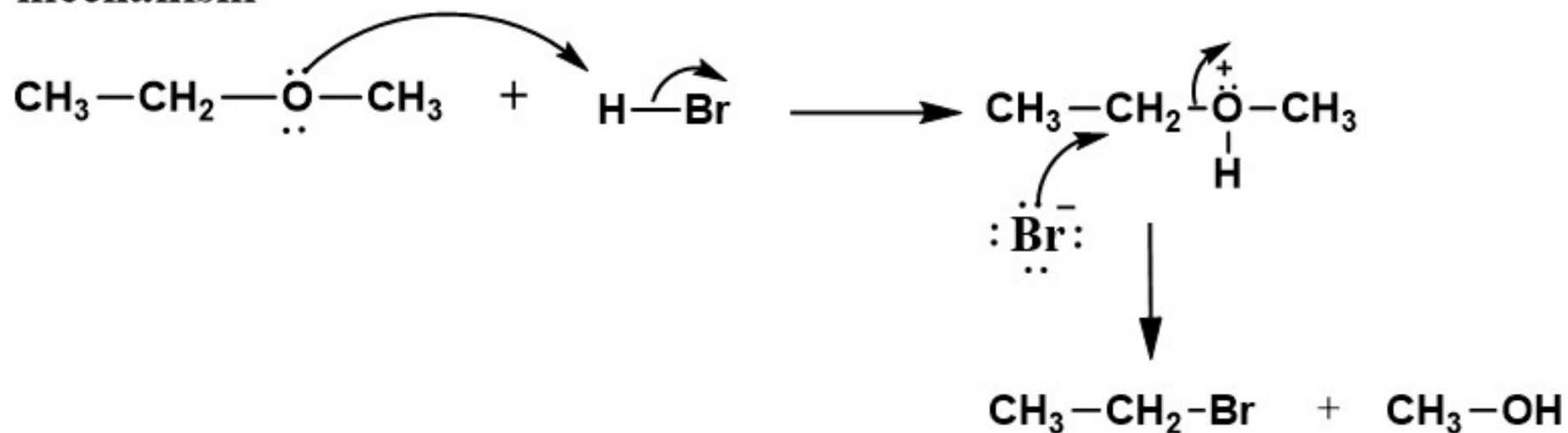
- Thionyl chloride (SOCl_2) and phosphorus halide (PX_3) are also used for this purpose



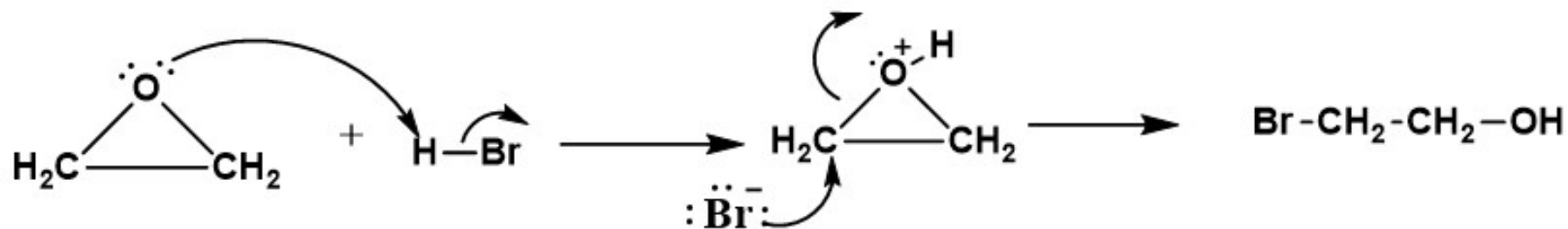
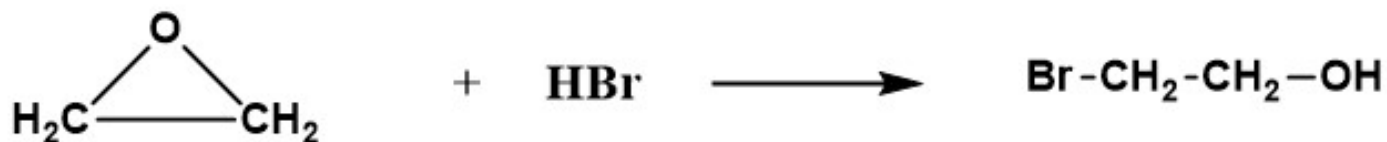
3. Ether



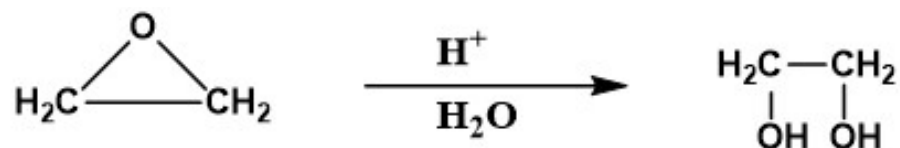
mechanism



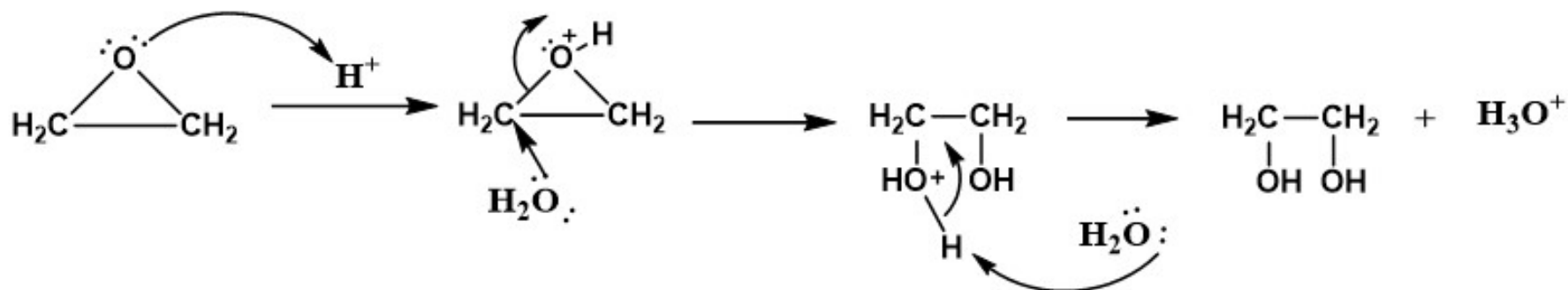
4. Epoxides



Acid catalysed cleavage of epoxides

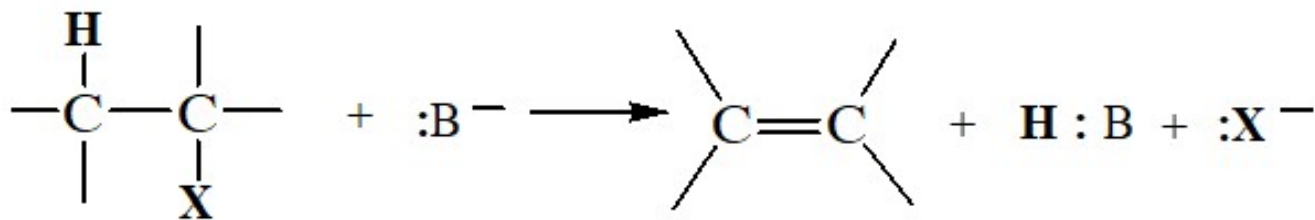


Mechanism

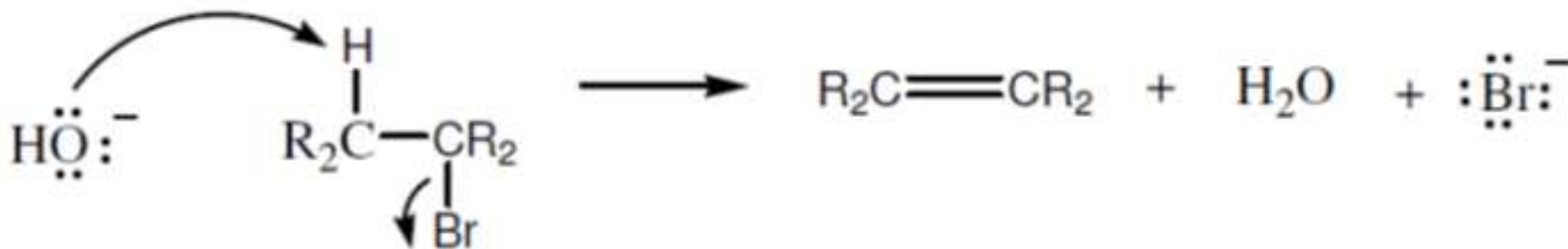


4.2 ELIMINATION REACTIONS

An elimination reaction is a type of organic reaction in which two substituents (two atoms or groups) are removed from a molecule



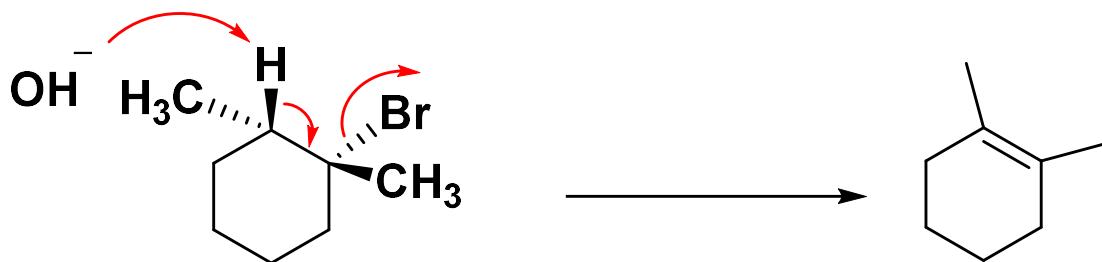
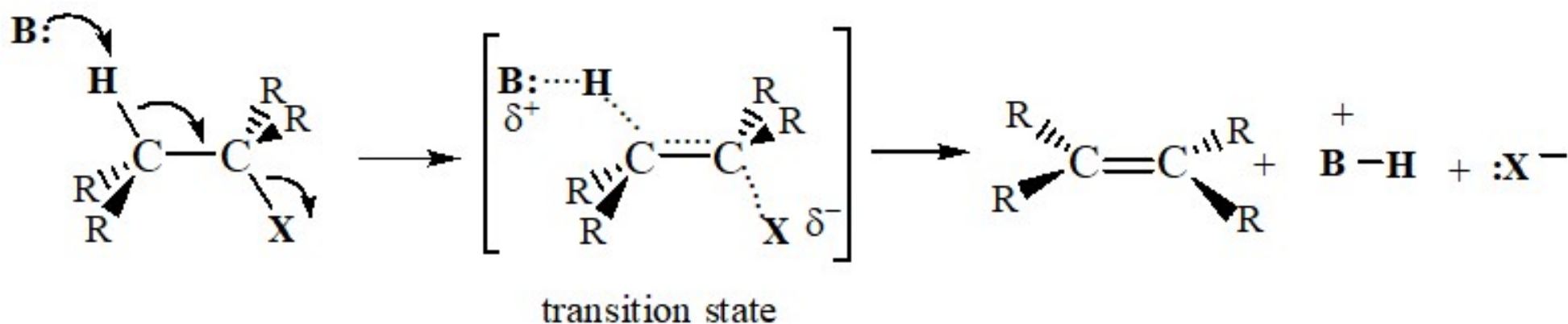
- base removes H^+ as X^- leaves



E2 and E1 Mechanism

E2 Elimination Mechanism

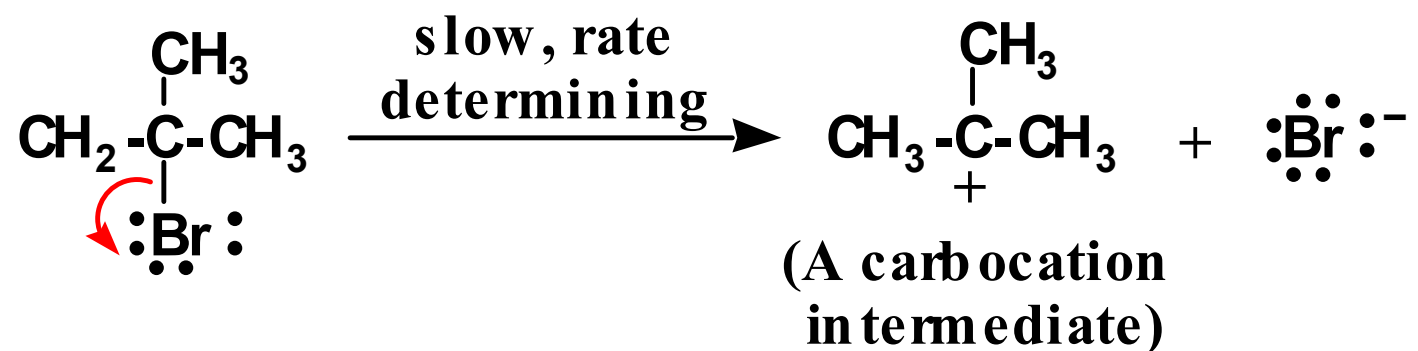
- concerted (single step)
- all bond-breaking and bond-forming steps are concerted
- the H and X eliminated must be aligned **anti** to one another



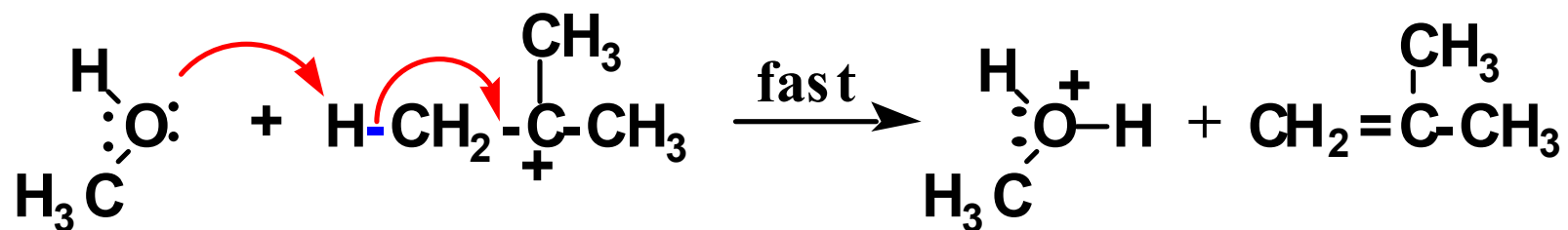
E1 Elimination Mechanism

- two-step mechanism

Step 1: ionization of C-X gives a carbocation intermediate

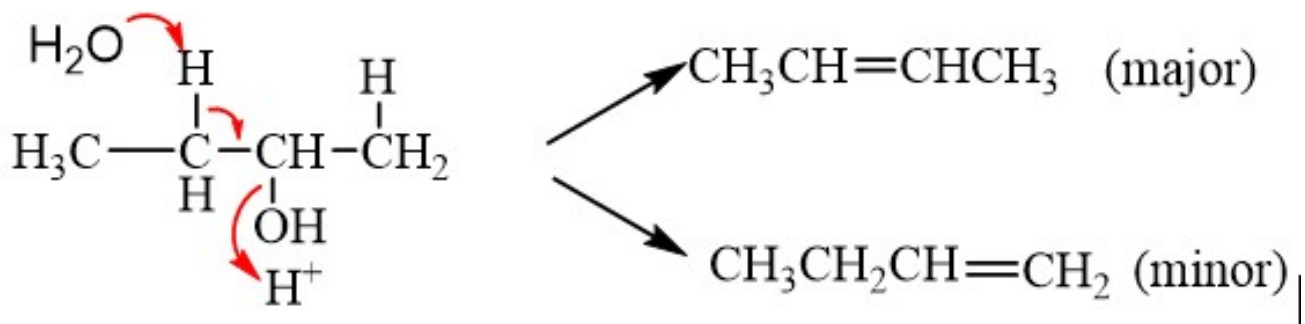


Step 2: proton transfer from the carbocation intermediate to a base (in this case, the solvent) gives the alkene

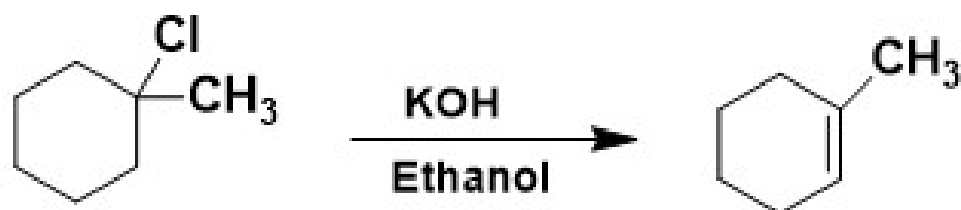
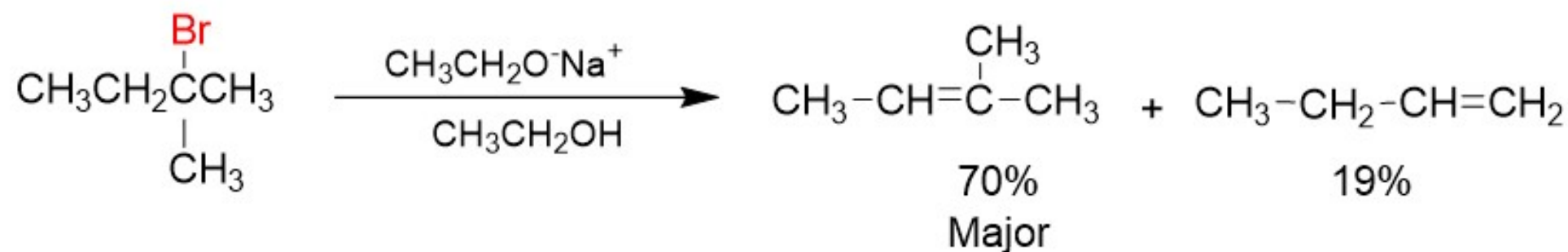
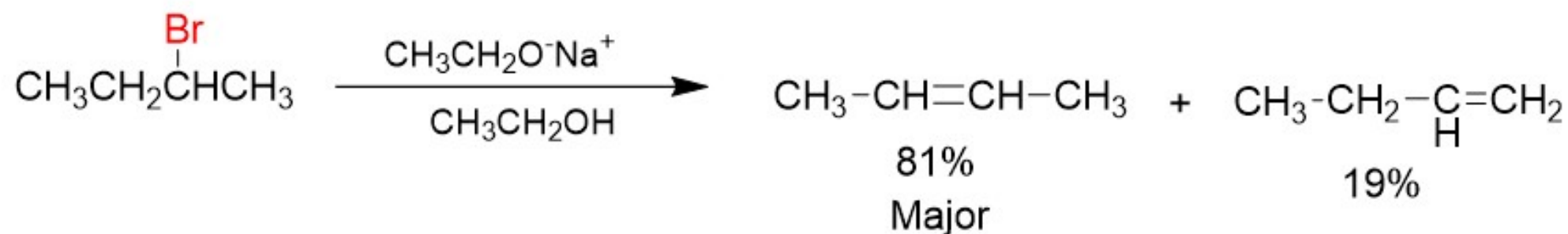


Zaitsev's Rule

- Elimination reactions almost always give mixtures of alkene products.
- Zaitsev's rule states that if more than one alkene can be formed by an elimination reaction, the **more stable alkene** is the major product.

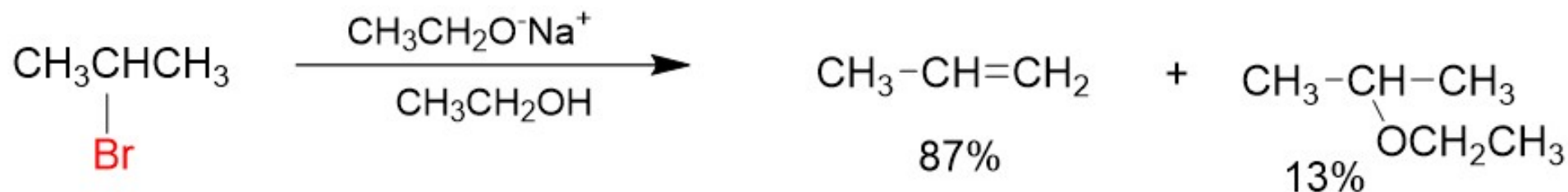
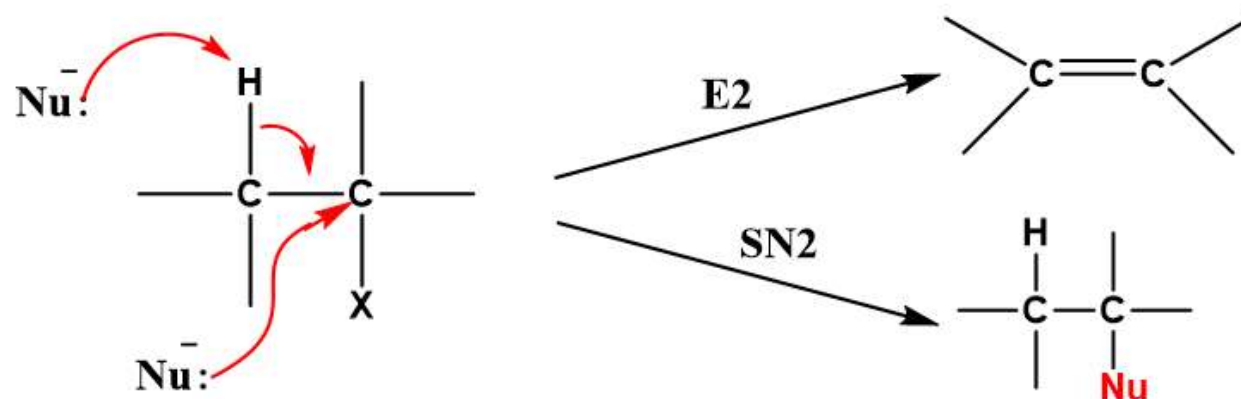


more highly substituted C=C double bond is more stable due to the electron donating properties of the alkyl group

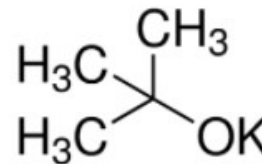


Elimination versus Substitution

Nucleophilic substitution and elimination reactions often compete. Since bases are nucleophilic, they can undergo substitution

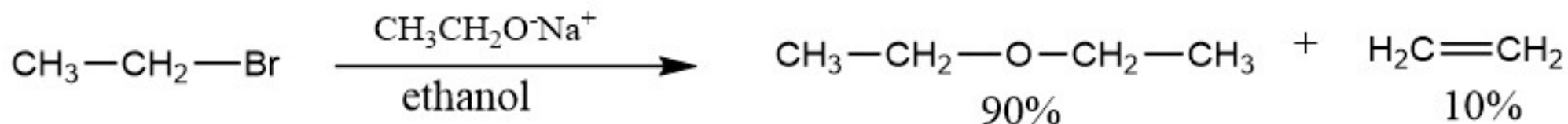


- **to favor elimination:** use a strong, hindered base
e.g., KOtBu (Potassium *tert*-butoxide)

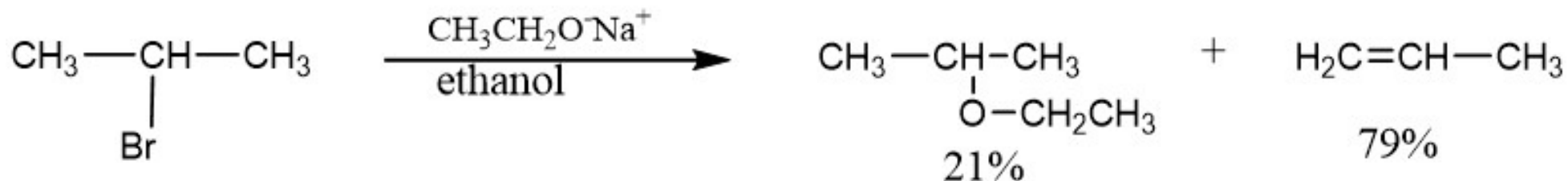


- **to favor substitution:** use a small, unhindered nucleophile
- **Reactivity Patterns**

With 1° halide substitution is highly favoured

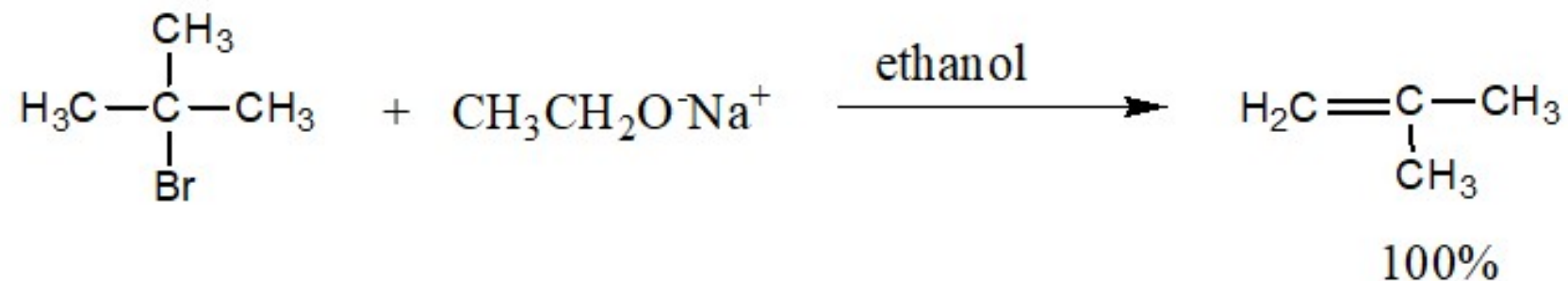


With 2° halide elimination favoured



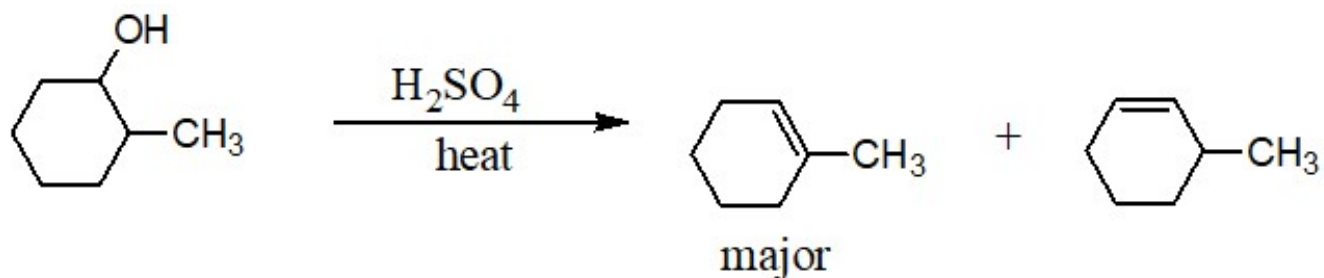
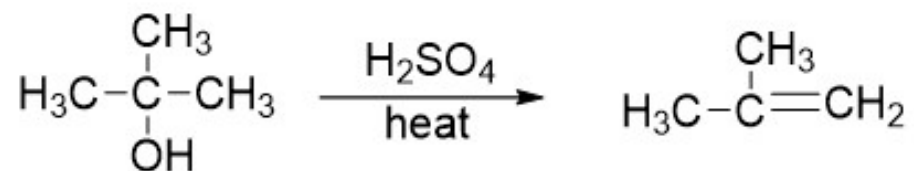
steric effect makes substitution difficult

With 3° halide elimination highly favoured

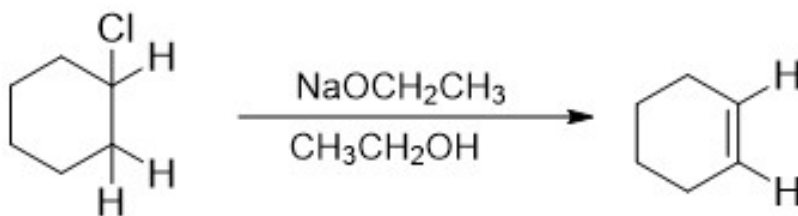
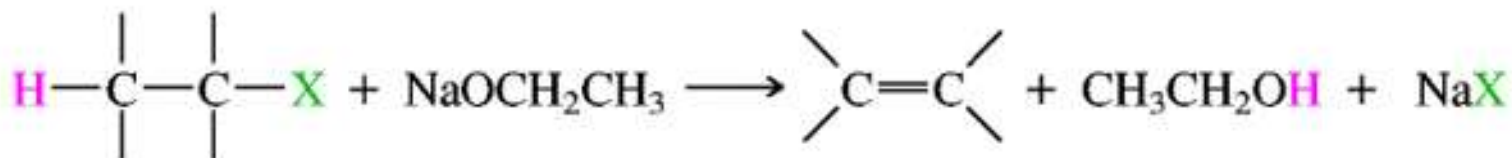
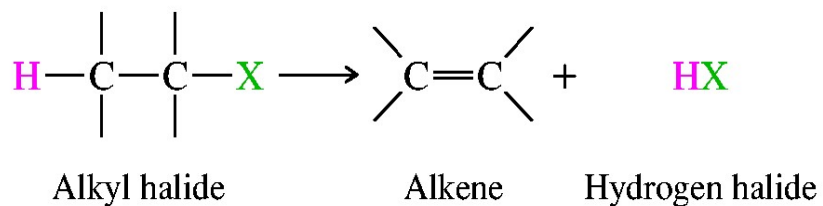


Applications of Elimination Reactions

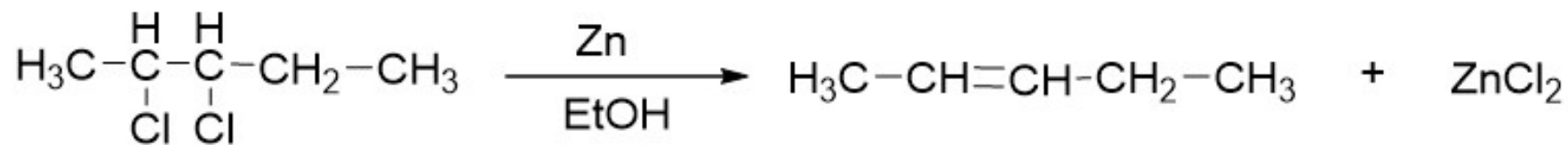
1. Dehydration of alcohol



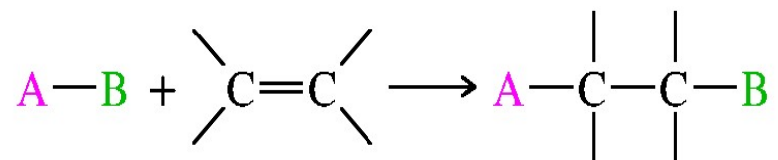
2. Dehydrohalogenation



3. Elimination of vicinal dihalides

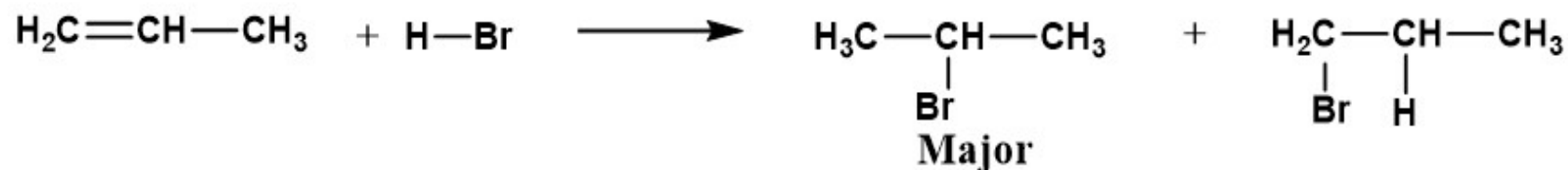


4.3 ADDITION REACTIONS

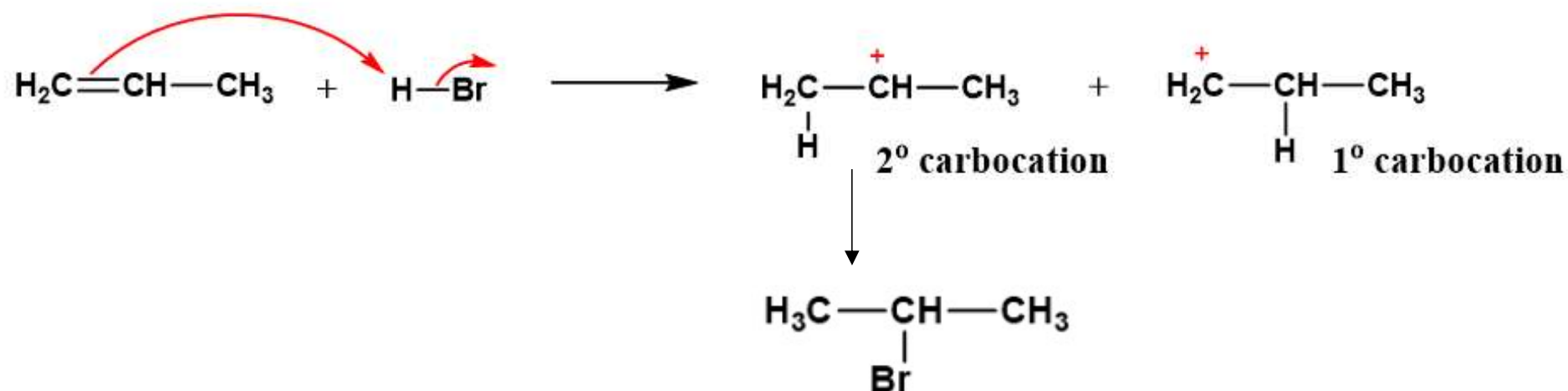


Rules for Addition Reactions

Markovnikov's Rule: addition reaction pass through the formation of more stable carbocation as an intermediate.



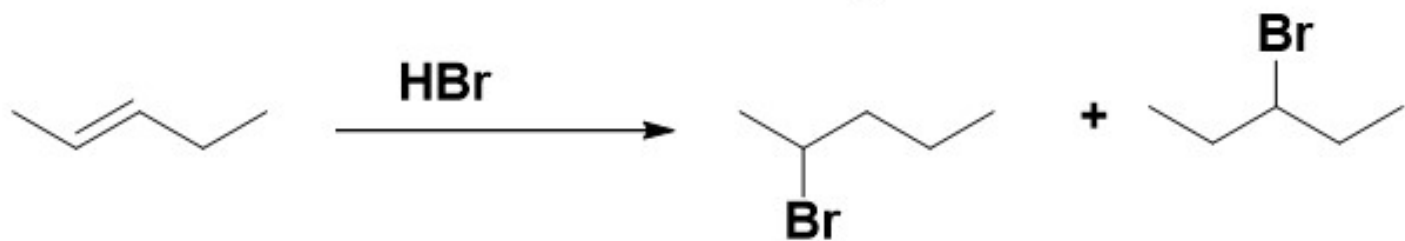
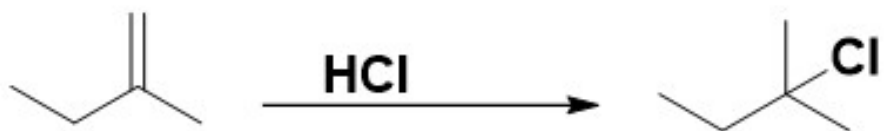
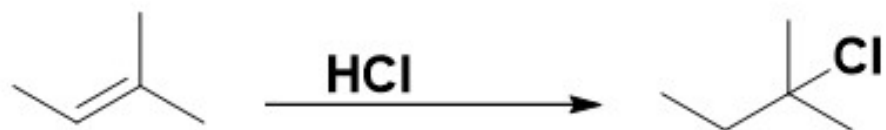
mechanism



The reaction tends to pass through more stable carbocation.

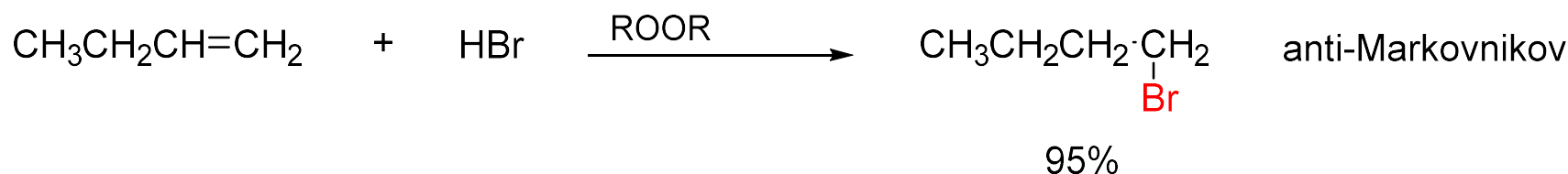
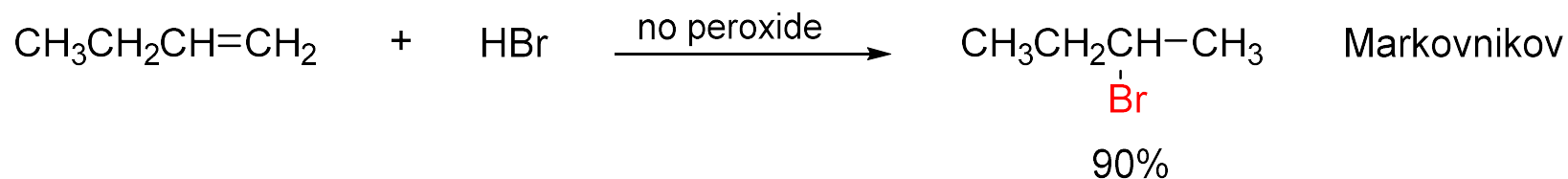
Carbocation stability: $3^\circ > 2^\circ > 1^\circ$

Examples

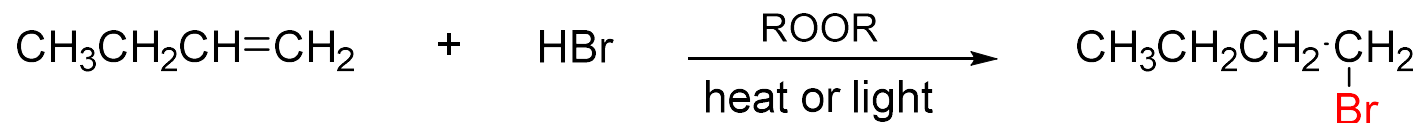


Anti-Markovnikov's (Radical) addition

Addition of HX on double bond in the presence of peroxides follow anti-Markovnikov's rule



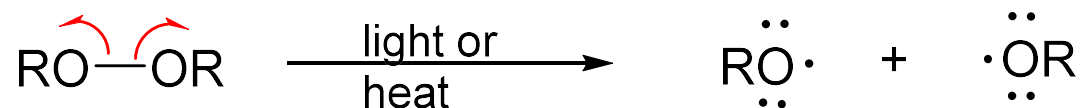
The overall reaction



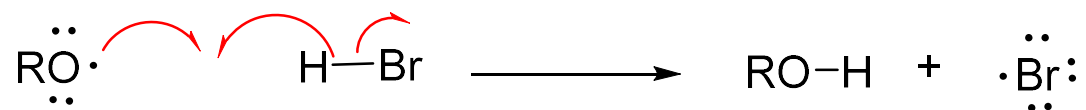
Mechanism

a) Initiation

Step 1 : dissociation of a peroxide in to two radicals

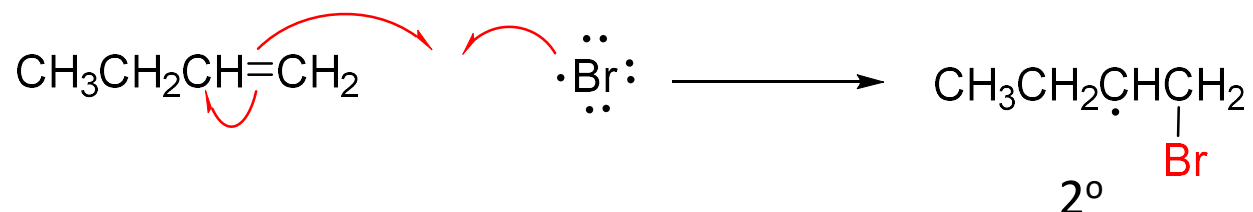


Step 2: Hydrogen atom abstraction from hydrogen bromide by an alkoxy radical:



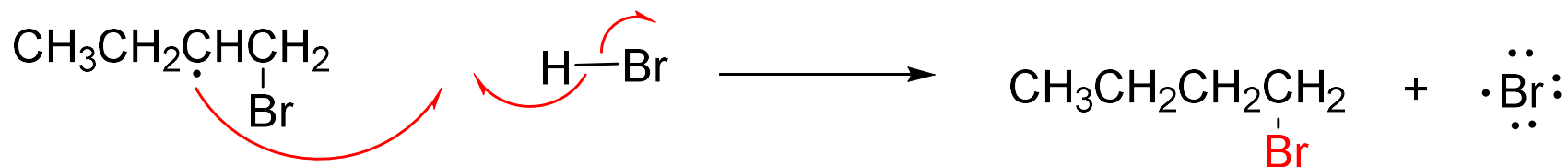
(b) Chain propagation

Step 3: Addition of a bromine atom to the alkene:



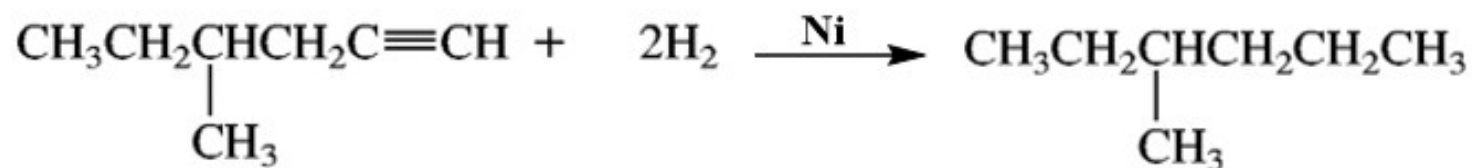
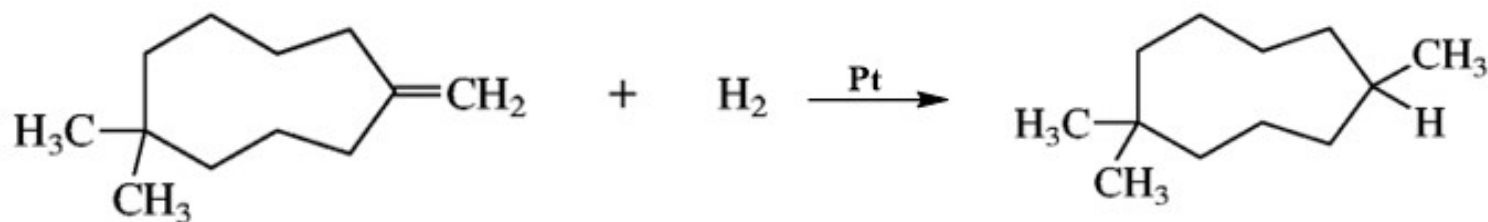
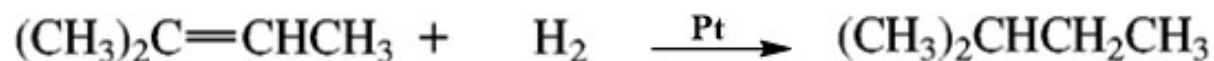
A secondary alkyl radical is more stable than a primary radical.

Step 4: Abstraction of a hydrogen atom from hydrogen bromide by the free radical formed in step 3:



Application of Addition Reactions

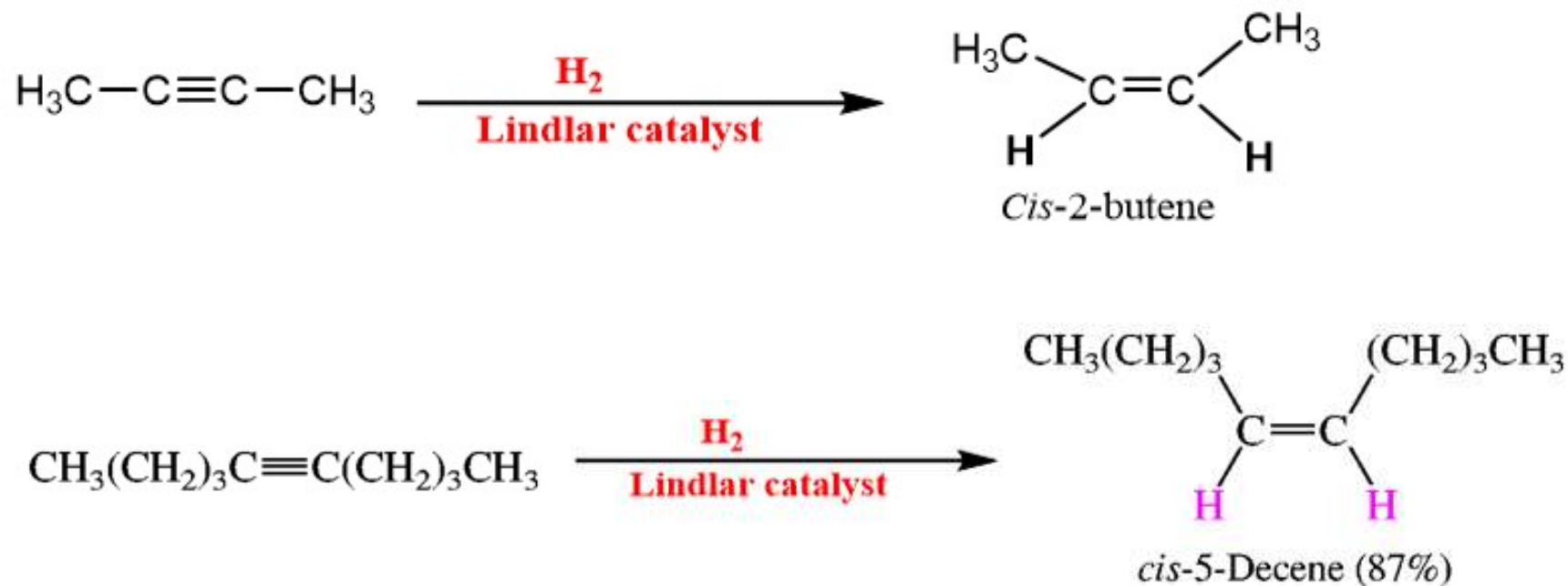
1. Hydrogenation



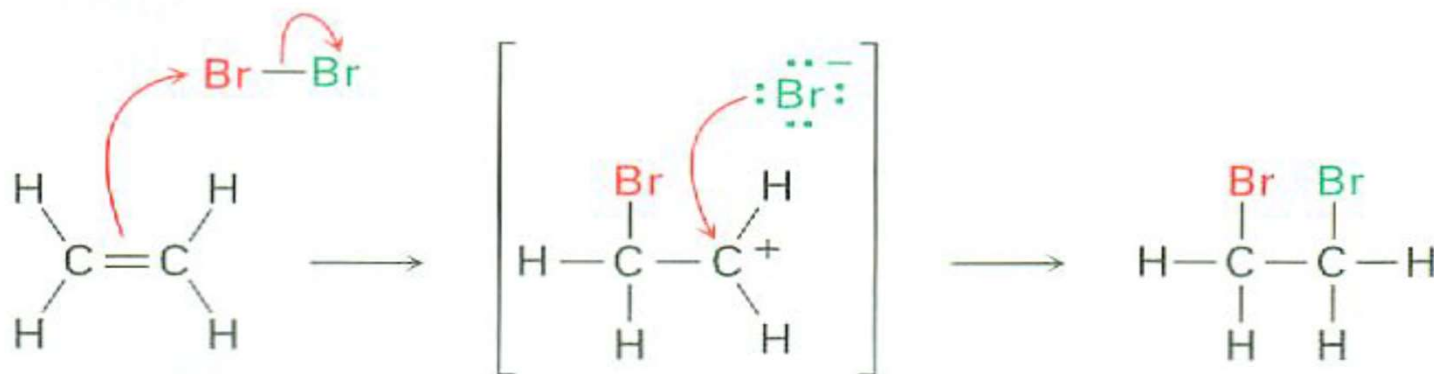
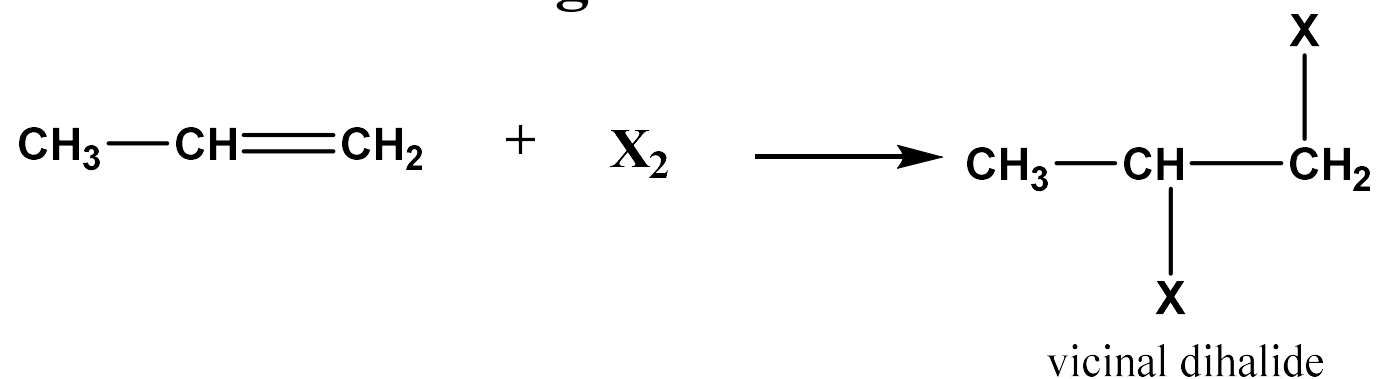
Partial hydrogenation of an alkyne to produce alkene could be achieved using Lindlar catalyst.

Lindlar catalyst = palladium on CaCO_3 combination to which lead acetate and quinoline have been added.

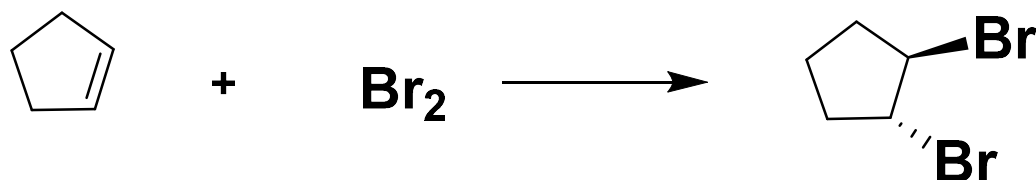
Hydrogenation of alkynes with internal triple bonds gives cis alkenes

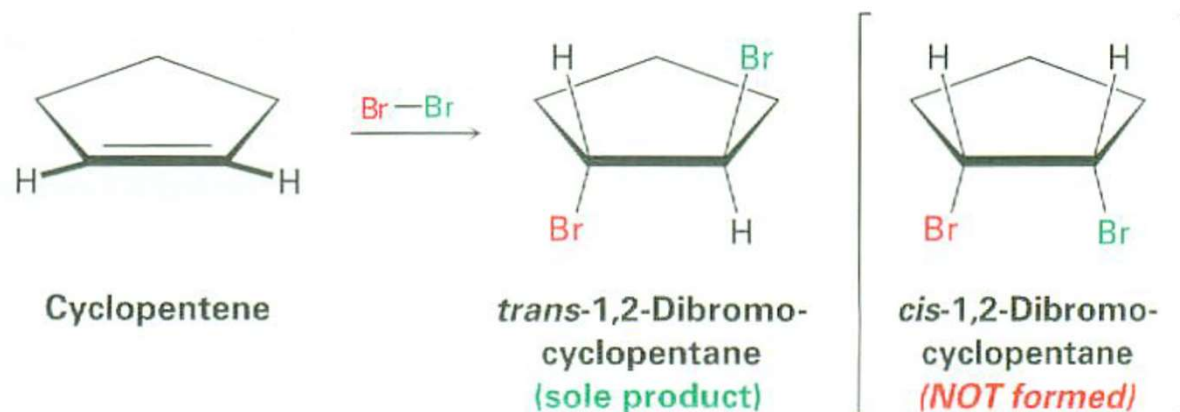


2. Addition of Halogens

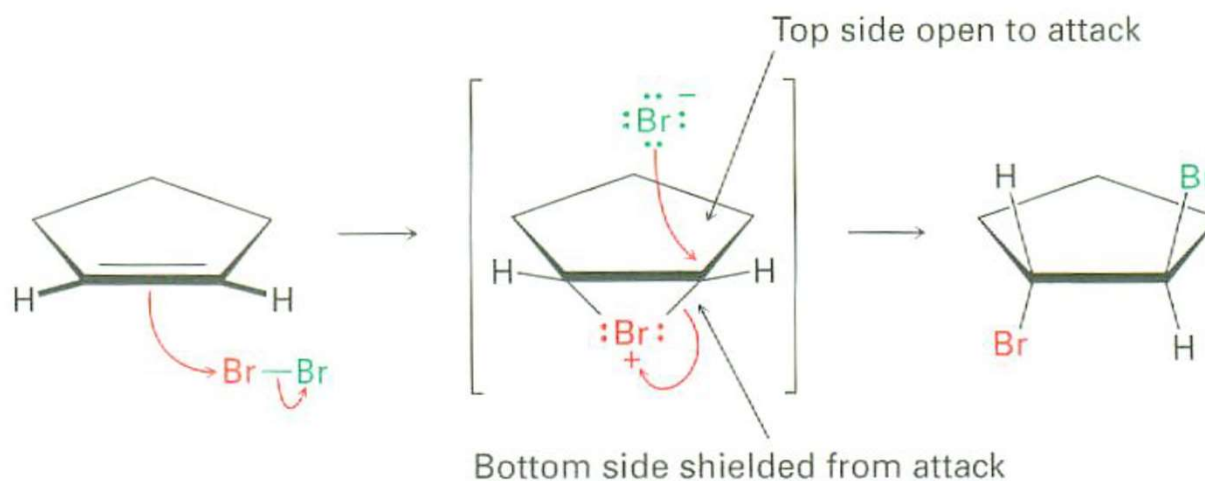
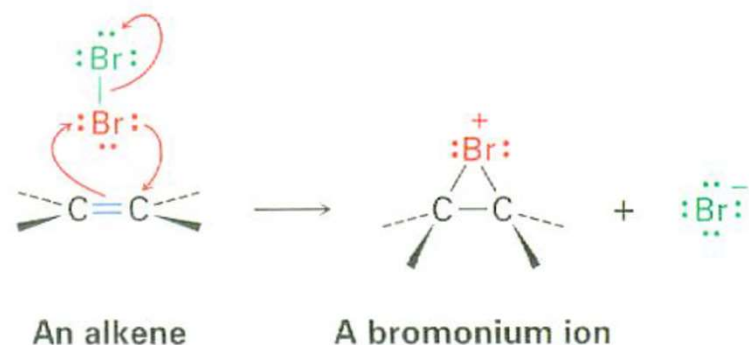


- anti-stereochemistry-two new groups are added to opposite sides of the original pi bond

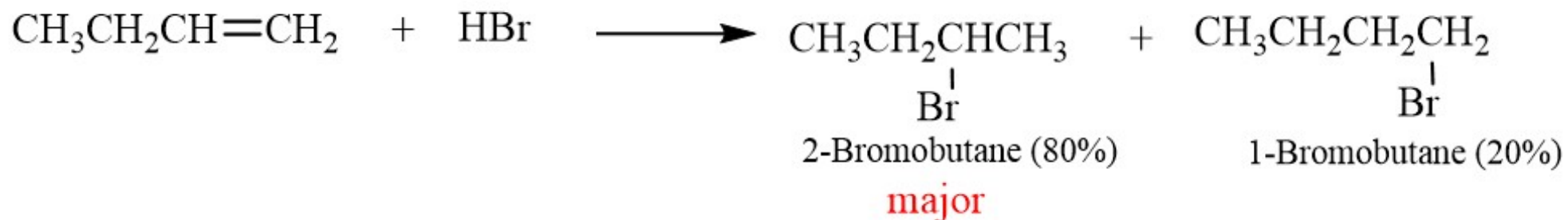




The reaction intermediate is not a carbocation but is instead a bromonium ion, R_2Br^+ . Formed by addition of Br^+ to the alkene.



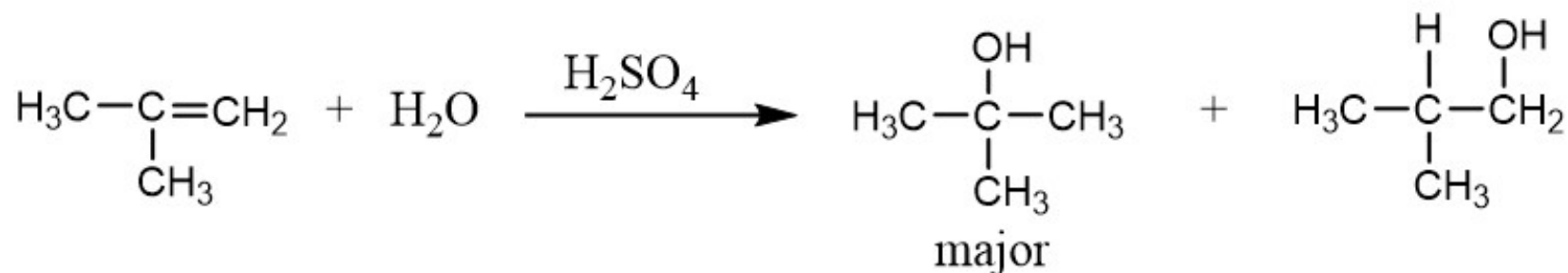
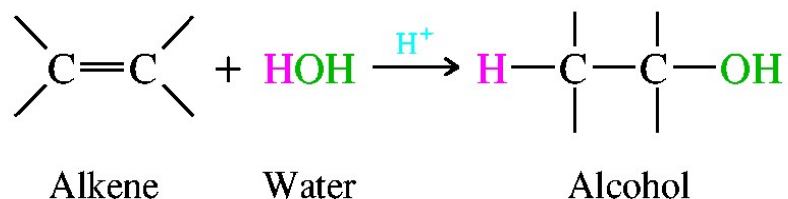
3. Addition of Hydrogen Halide (HX)



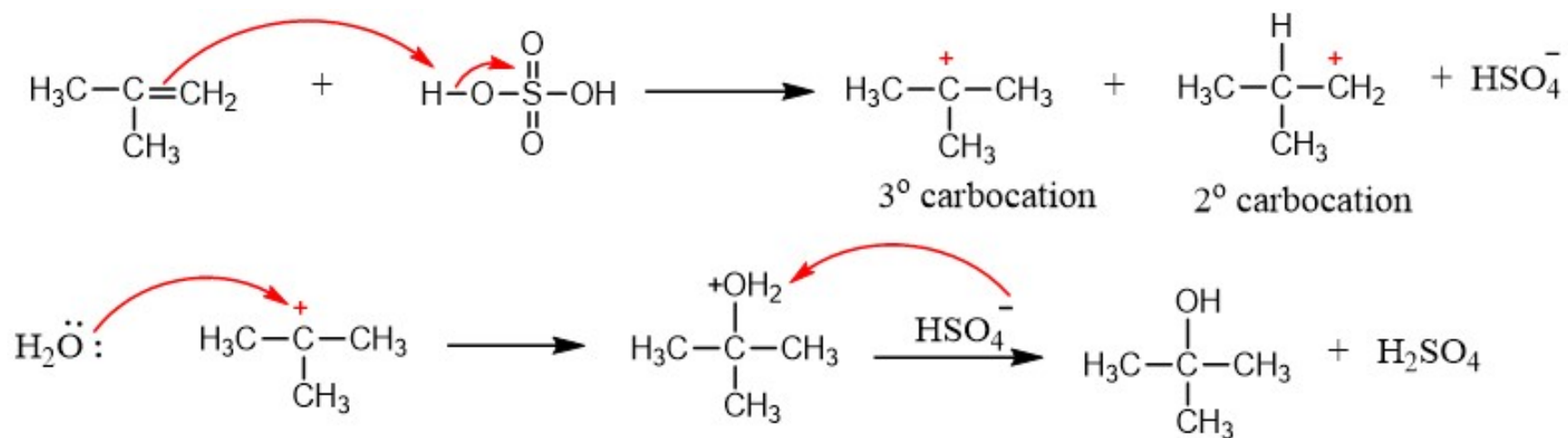
The major product is determined by Markovnikov's rule



4. Addition of water (Hydration)

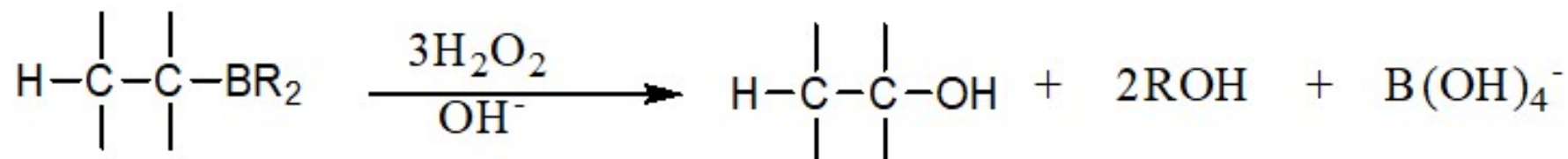
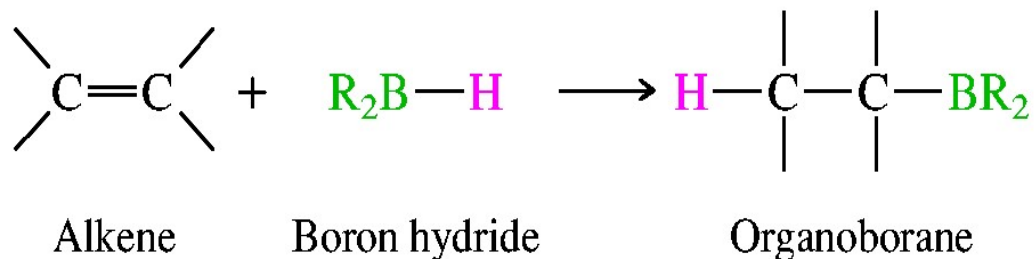


The reaction follows Markovnikov's rule.

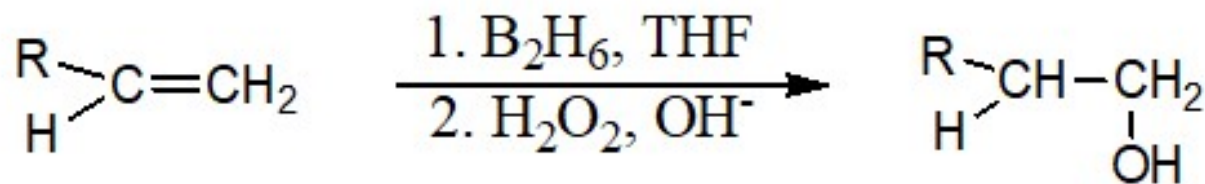


5. Hydroboration-Oxidation of Alkenes

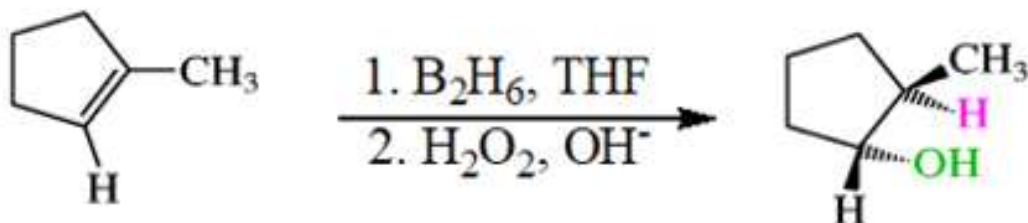
- A two steps reaction



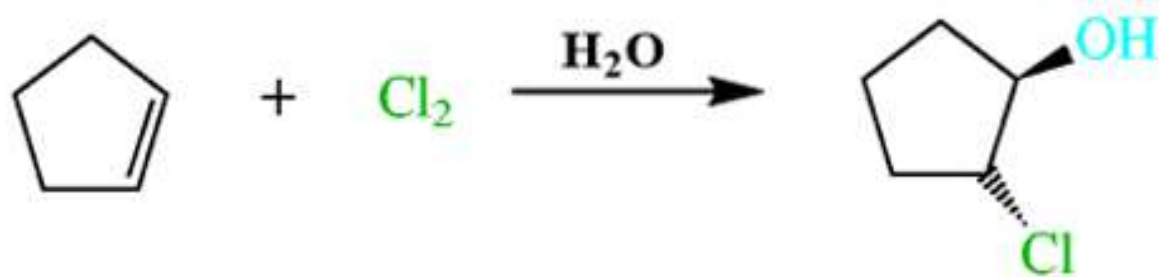
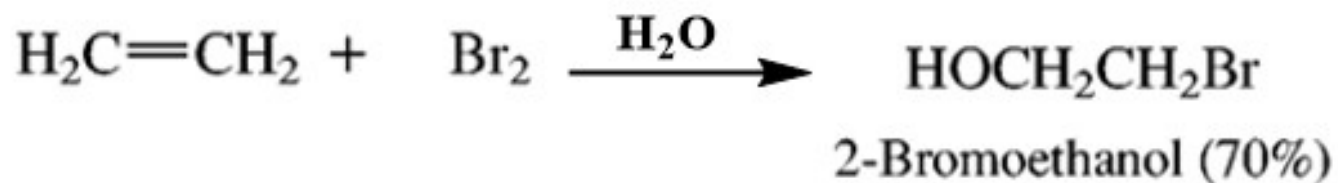
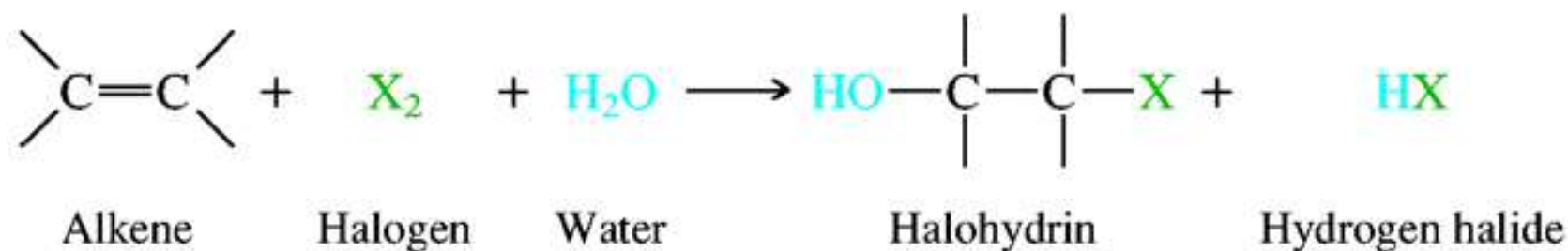
- borane-tetrahydrofuran complex ($\text{H}_3\text{B}-\text{THF}$)-common hydroborating agent



anti-Markovnikov's
addition

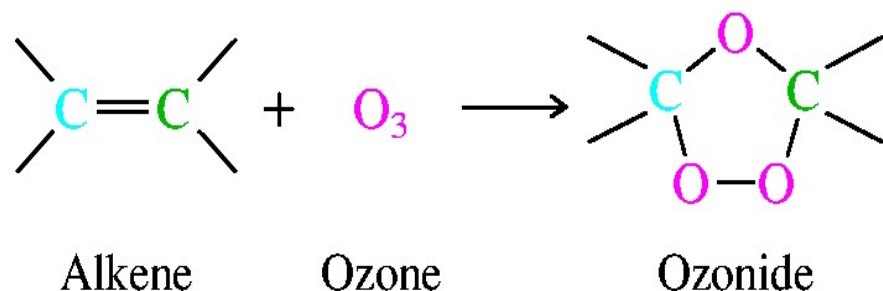


6. Conversion of Alkenes to Vicinal Halohydrins

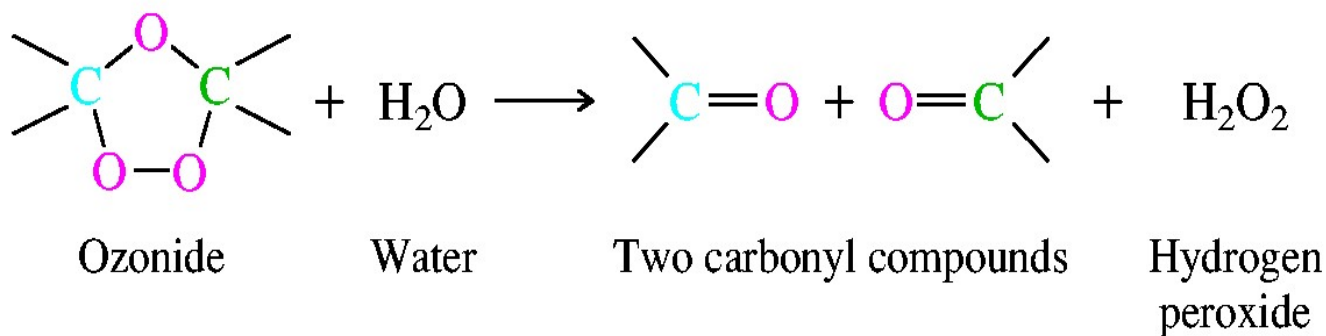


7. Ozonolysis of Alkenes

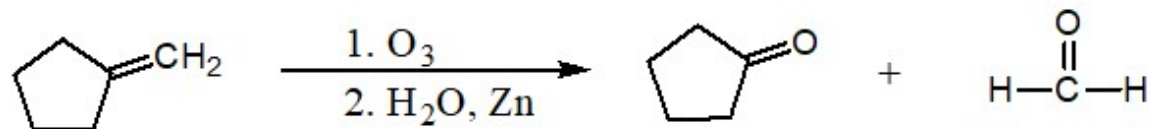
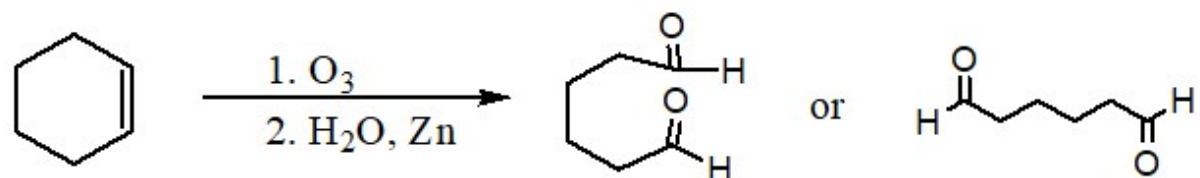
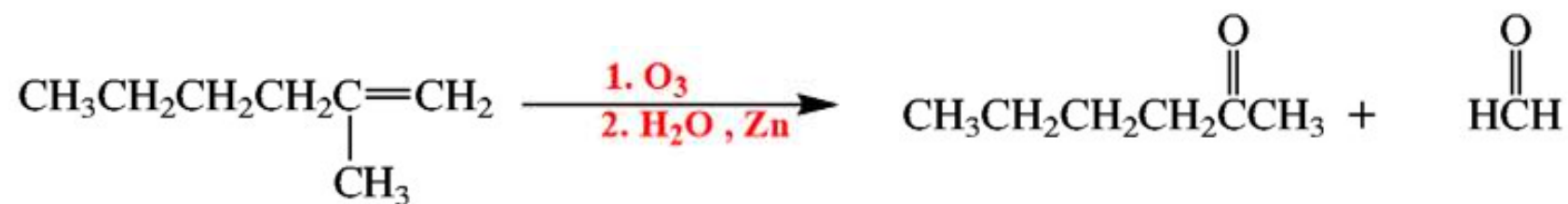
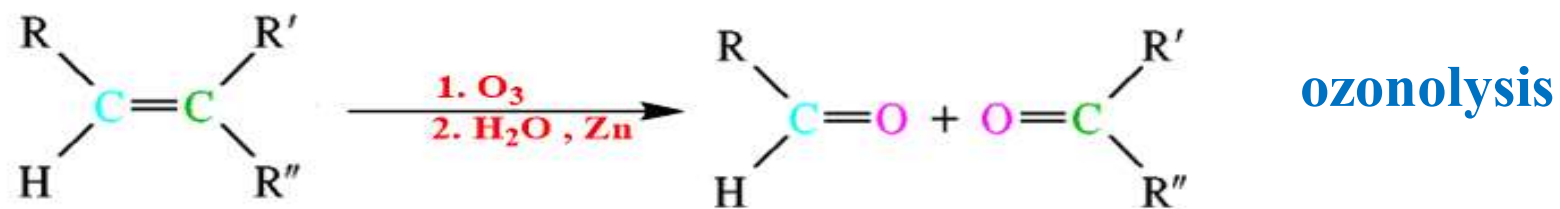
Ozone is a powerful electrophile and undergoes a remarkable reaction with alkenes in which both the σ and π components of the carbon-carbon double bond are cleaved to give a product referred to as an **ozonide**.



Ozonides undergo hydrolysis in water, giving carbonyl compounds.

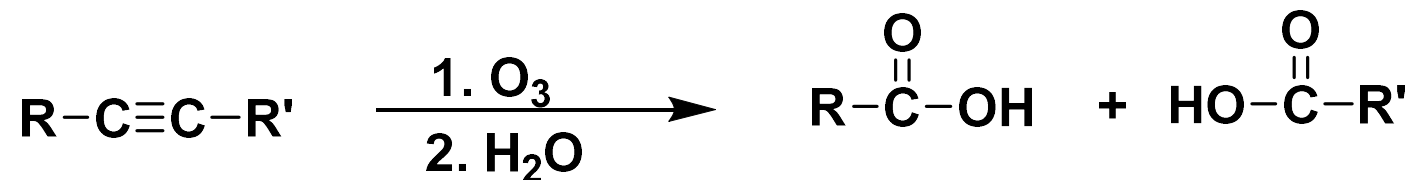


The two-stage reaction sequence is called **ozonolysis**

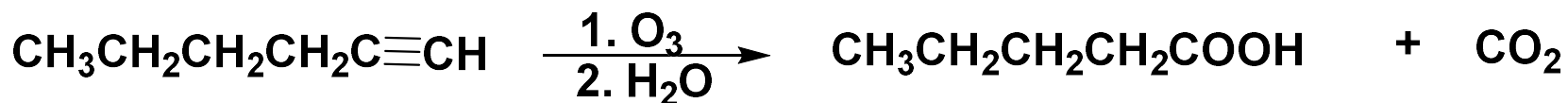
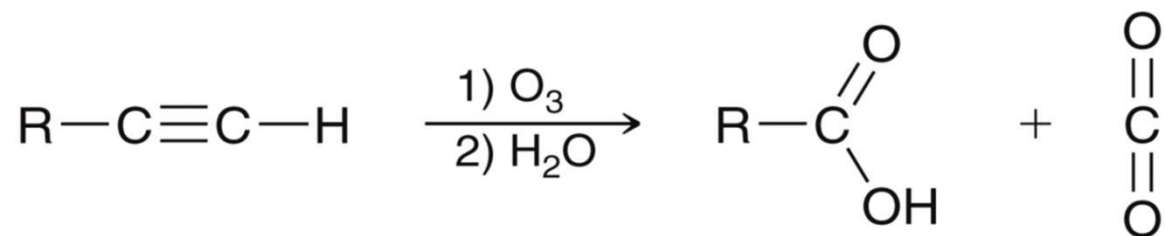


8. Ozonolysis of Alkynes

- **Ozonolysis** of an **internal alkyne** produces **two carboxylic acids**

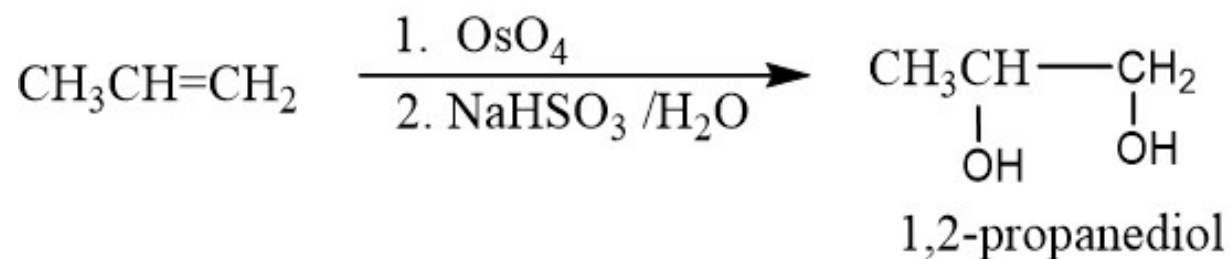
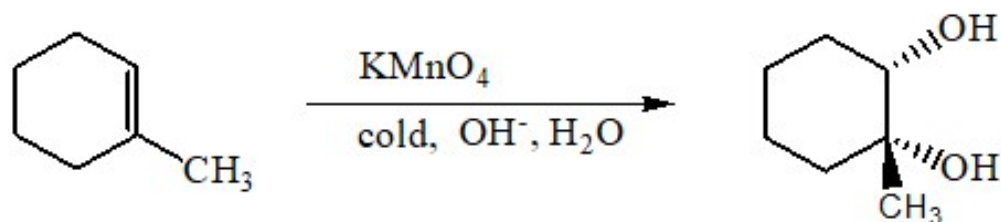
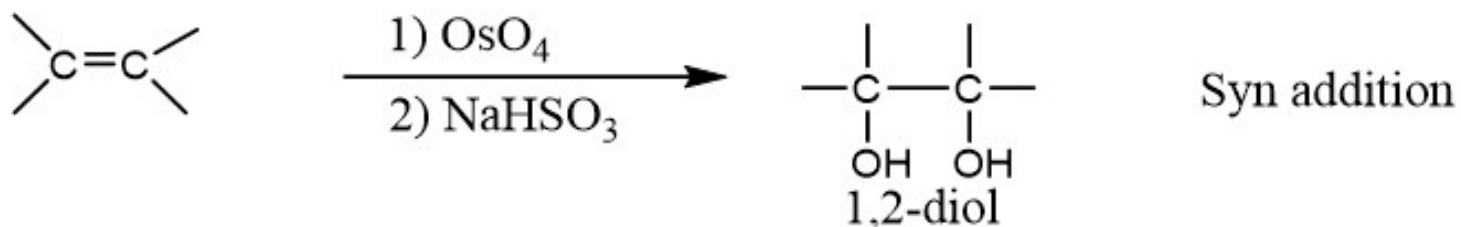
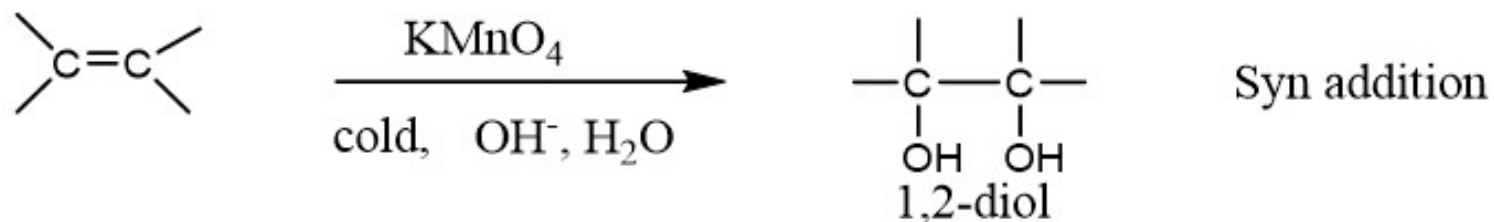


- **Ozonolysis** of a **terminal alkyne** yields a **carboxylic acid and carbon dioxide**.



8. Hydroxylation

- addition of two OH groups to each of the two double bonded carbon



9. Oxidative cleavage of an alkene using hot alkaline KMnO_4

Acidic KMnO_4 also causes double bond cleavage like ozone.

